

Nazariy savol 1

Tushayotgan yuk bilan mayatnik

Radiusi R bo'lgan qattiq silindrsimon (doira shaklidagi kesimga ega) sterjen yer ustida gorizontol holda mahkamlangan. Massasi e'tiborga olinmaydigan va uzunligi L ($L > 2\pi R$) bo'lgan ip yordamida sterjenning tepa qismidagi A nuqtadan massasi m bo'lgan mayatnik yuki osilgan (1a-rasm ko'rsatilgandek). Bob (mayatnik yuki) A nuqta bilan bir xil balandlikka ko'tariladi, so'ng ip tarang holatda tinchlikdan qo'yib yuboriladi. Ipning cho'zilishini hisobga olmang. Mayatnik yukini massaviy nuqta deb qarash mumkin va u faqat sterjen o'qiga perpendikulyar (tik) tekislikda tebranadi. Shunga ko'ra, mayatnik yuki bundan buyon *zarracha* deb ataladi. Erkin tushish tezlanishi $\bar{g}$ ga teng.

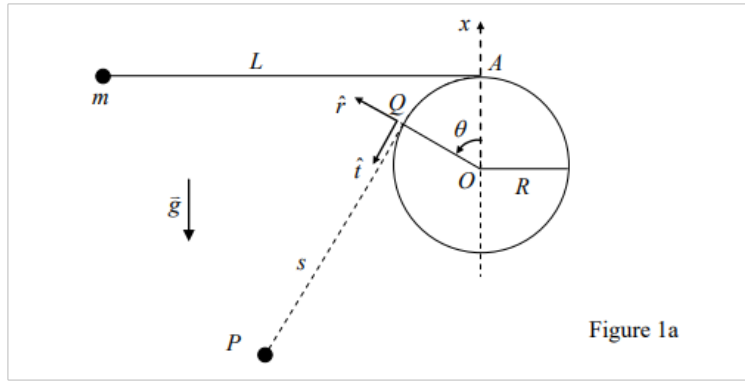


Figure 1: 1a-rasm

O nuqta koordinatalar sistemasining boshi bo'lsin. Zarracha P nuqtada bo'lganda, ip silindr sirtiga Q nuqtada urinma bo'ladi. QP kesmaning uzunligi s deb belgilanadi. Q nuqtadagi birlik urinma vektor (aylanaga urinma yo'nalishdagi birlik vektor) va birlik radial vektor (markazdan chiquvchi birlik vektor) mos ravishda $\hat{t}$ va $\hat{r}$ orqali berilgan. OQ radiusning burchak siljishi θ , OA bo'ylab vertikal x o'qidan soat miliga teskari yo'nalishda o'lchanganda, musbat qabul qilinadi.

$\theta = 0$ bo'lganda, s uzunlik L ga teng va zarrachaning gravitatsion potensial energiyasi (vaziyat energiyasi) U nolga teng. Zarracha harakatlanishi davomida θ va s ning oniy o'zgarish tezliklari (vaqt bo'yicha hosilalari) mos ravishda $\dot{\theta}$ va $\dot{s}$ bilan belgilanadi.

Agar boshqacha ko'rsatilmagan bo'lsa, barcha tezlik va tezlik vektorlari qo'zg'almas O nuqtaga nisbatan olingan.

A qism

A qismda zarracha harakatlanayotganda ip tarang bo'ladi. Yuqorida kiritilgan kattaliklar (ya'ni s , θ , $\dot{s}$, $\dot{\theta}$, R , L , g , $\hat{t}$ va $\hat{r}$) orqali quyidagilarni toping:

- $\dot{\theta}$ va $\dot{s}$ orasidagi munosabat. (0.5 ball)
- Harakatlanuvchi Q nuqtaning O ga nisbatan tezligi $\vec{v}_Q$. (0.5 ball)
- Zarracha P da bo'lganda uning harakatlanuvchi Q nuqtaga nisbatan tezligi $\vec{v}'$. (0.7 ball)

- (d) Zarracha P da bo'lganda uning O ga nisbatan tezligi $\vec{v}$. (0.7 ball)
- (e) Zarracha P da bo'lganda uning O ga nisbatan *tezlanishining* $\hat{t}$ tashkil etuvchisi. (0.7 ball)
- (f) Zarracha P da bo'lganda uning gravitatsion potensial energiyasi U . (0.5 ball)
- (g) Zarrachaning traektoriyasining eng past nuqtasidagi tezligi v_m . (0.7 ball)

B qism

B qismda L ning R ga nisbati quyidagi qiymatga ega:

$$\frac{L}{R} = \frac{9\pi}{8} + \frac{2}{3} \cot \frac{\pi}{16} = 3.534 + 3.352 = 6.886$$

- (h) Q dan P gacha bo'lgan ip qismi ham to'g'ri, ham eng qisqa bo'lganda zarrachaning tezligi v_s qanday? (g va R orqali ifodalang) (2.4 ball)
- (h) Zarracha sterjenning narigi tomoniga tebranib o'tganda eng yuqori H nuqtadagi tezligi v_H qanday? (g va R orqali ifodalang) (1.9 ball)

C qism

C qismda, mayatnik yuki (massasi m) A dan osilish o'rniga, ip sterjenning tepasidan oshirilib, massasi M bo'lgan og'irroq yukka ulangan, 1b-rasmda ko'rsatilgandek. Yuk ham zarracha deb qaralishi mumkin.

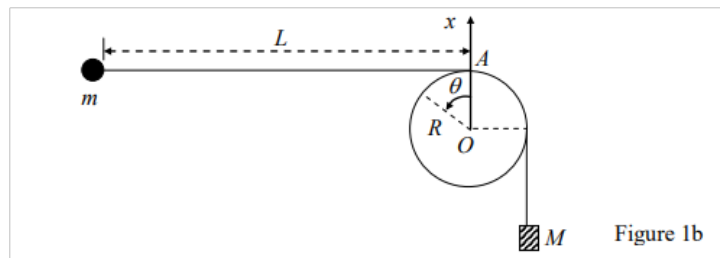


Figure 2: 1b-rasm

Dastlab, mayatnik yuki A bilan bir xil balandlikda ushlab turiladi, shunda og'irlik O dan pastda osilib turganda, ip tarang va uning gorizontal qismining uzunligi L ga teng bo'ladi. So'ng mayatnik yuki tinchlikdan qo'yib yuboriladi va yuk tusha boshlaydi. Mayatnik yuki vertikal tekislikda qoladi va tushayotgan yukning yonidan hech qanday to'siqsiz tebranib o'ta oladi, deb faraz qiling.

Ip bilan sterjen sirti orasidagi kinetik ishqalish (harakatdagi ishqalish) e'tiborga olinmaydi. Biroq, statik ishqalish (tinchlikdagi ishqalish) yetarlicha katta deb faraz qilinadi, shundayki, yuk to'xtagandan so'ng (ya'ni tezligi nolga teng bo'lgach) u qo'zg'almasdan qoladi.

- (j) Faraz qilaylik, yuk haqiqatan ham D masofaga tushib to'xtaydi va $(L - D) \gg R$ shart bajariladi. Agar shundan keyin zarracha sterjen atrofida $\theta = 2\pi$ ga tebranib, ipning

sterjenga tegmaydigan ikkala qismi ham to'g'ri holda qolsa, $\alpha = D/L$ nisbat kritik qiymat α_c dan kichik bo'lmasligi kerak. R/L tartibidagi yoki undan yuqori hadlarni hisobga olmay, α_c ni M/m orqali baholang. (3.4 ball)

(a) $\dot{\theta}$ va $\dot{s}$ orasidagi munosabat:

(b) Harakatlanuvchi Q nuqtaning O ga nisbatan tezligi:

$$\vec{v}_Q =$$

(c) P da bo'lganda zarrachaning harakatlanuvchi Q ga nisbatan tezligi:

$$\vec{v}' =$$

(d) P da bo'lganda zarrachaning O ga nisbatan tezligi:

$$\vec{v} =$$

(e) P da bo'lganda zarracha tezlanishining $\hat{t}$ tashkil etuvchisi:

(f) P da bo'lganda zarrachaning gravitatsion potensial energiyasi:

$$U =$$

(g) Traektoriyaning eng past nuqtasidagi zarracha tezligi:

$$v_m =$$

(h) QP kesmasi to'g'ri va eng qisqa bo'lganda zarracha tezligi (g va R orqali ifoda va qiymat):

$$v_s =$$

(i) Eng yuqori nuqtadagi tezlik (g va R orqali ifoda va qiymat):

$$v_H =$$

(j) M/m massa nisbati orqali D/L nisbatning kritik qiymati α_c :

$$\alpha_c =$$

Nazariy masala 1. Yechim

Og'irlik tushayotgan belanchak

A qism

- (a) (a ball) Ipning uzunligi $L = s + R\theta$ o'zgarimas bo'lgani uchun, uning o'zgarish tezligi nolga teng bo'lishi kerak. Shuning uchun bizda:

$$\dot{s} + R\dot{\theta} = 0 \quad (A1)$$

- (b) O ga nisbatan Q nuqtasi R radiusli aylana bo'ylab burchak tezligi $\dot{\theta}$ bilan harakatlanaadi, shuning uchun:

$$\vec{v}_Q = R\dot{\theta}\hat{t} = -\dot{s}\hat{t} \quad (A2)$$

- (c) A1-rasmga murojaat qiling. Q ga nisbatan P nuqtasining Δt vaqt oralig'idagi siljishi:

$$\Delta\vec{r}' = (s\Delta\theta)(-\hat{r}) + (\Delta s)\hat{t} = [(s\dot{\theta})(-\hat{r}) + \dot{s}\hat{t}]\Delta t.$$

Bundan kelib chiqadi:

$$\vec{v}' = -s\dot{\theta}\hat{r} + \dot{s}\hat{t} \quad (A3)$$

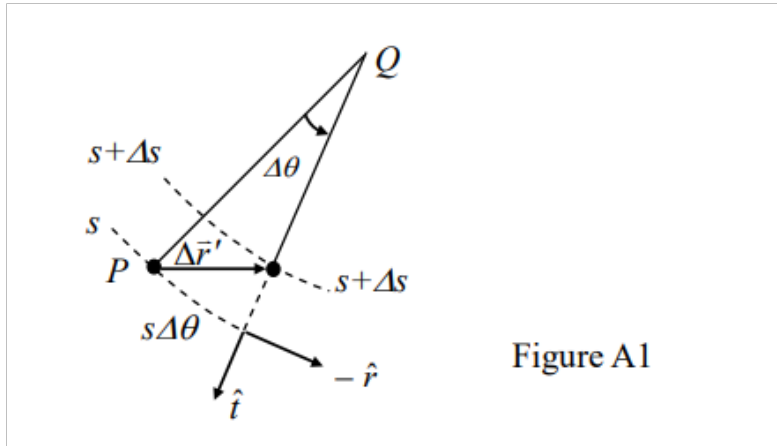


Figure 1: 1a-rasm (A1-rasm)

- (d) Zarrachaning O ga nisbatan tezligi (A2) va (A3) tenglamalarida berilgan ikki nisbiy tezlikning yig'indisiga teng:

$$\vec{v} = \vec{v}' + \vec{v}_Q = (-s\dot{\theta}\hat{r} + \dot{s}\hat{t}) + R\dot{\theta}\hat{t} = -s\dot{\theta}\hat{r} \quad (A4)$$

- (e) A2-rasmga murojaat qiling. Tezlik o'zgarishining $\Delta\vec{v}$ ($-\hat{t}$)-tashkil etuvchisi:

$$(-\hat{t}) \cdot \Delta\vec{v} = v\Delta\theta = v\dot{\theta}\Delta t.$$

Shuning uchun tezlanishning $\vec{a} = \Delta\vec{v}/\Delta t$ $\hat{t}$ -tashkil etuvchisi:

$$\hat{t} \cdot \vec{a} = -v\dot{\theta}.$$

(A4) tenglamaga ko'ra zarrachaning tezligi $v = s\dot{\theta}$ bo'lgani uchun, P nuqtadagi zarracha tezlanishining $\hat{t}$ -tashkil etuvchisi:

$$\vec{a} \cdot \hat{t} = -v\dot{\theta} = -(s\dot{\theta})\dot{\theta} = -s\dot{\theta}^2 \quad (\text{A5})$$

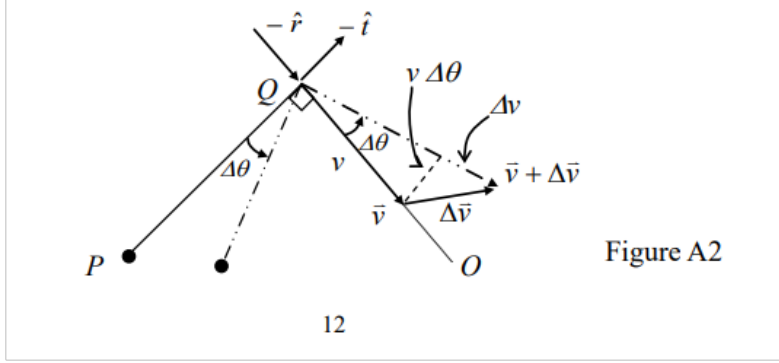


Figure 2: 1b-rasm (A2-rasm)

Nazariy masala 1. Yechim

Og'irlik tushayotgan belanchak

A qism

- (a) (a ball) Ipning uzunligi $L = s + R\theta$ o'zgaras bo'lgani uchun, uning o'zgarish tezligi nolga teng bo'lishi kerak. Shuning uchun bizda:

$$\dot{s} + R\dot{\theta} = 0 \quad (\text{A1})$$

- (b) O ga nisbatan Q nuqtasi R radiusli aylana bo'ylab burchak tezligi $\dot{\theta}$ bilan harakatlanadi, shuning uchun:

$$\vec{v}_Q = R\dot{\theta}\hat{t} = -\dot{s}\hat{t} \quad (\text{A2})$$

- (c) A1-rasmga murojaat qiling. Q ga nisbatan P nuqtasining Δt vaqt oralig'idagi siljishi:

$$\Delta \vec{r}' = (s\Delta\theta)(-\hat{r}) + (\Delta s)\hat{t} = [(s\dot{\theta})(-\hat{r}) + \dot{s}\hat{t}]\Delta t.$$

Bundan kelib chiqadi:

$$\vec{v}' = -s\dot{\theta}\hat{r} + \dot{s}\hat{t} \quad (\text{A3})$$

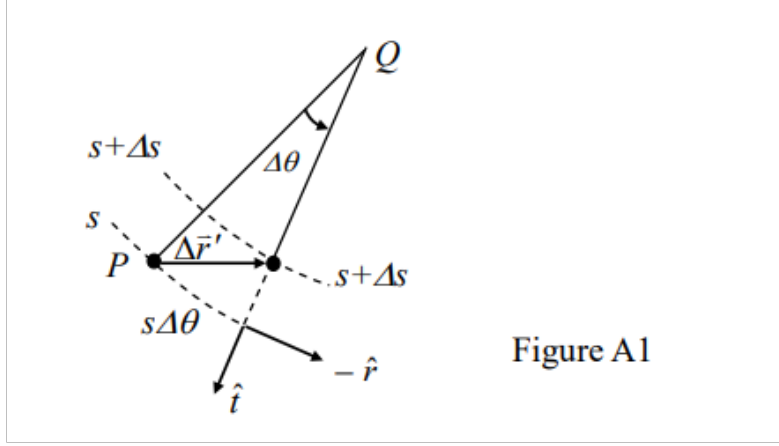


Figure 3: 1a-rasm (A1-rasm)

- (d) Zarrachaning O ga nisbatan tezligi (A2) va (A3) tenglamalarida berilgan ikki nisbiy tezlikning yig'indisiga teng:

$$\vec{v} = \vec{v}' + \vec{v}_Q = (-s\dot{\theta} \hat{r} + \dot{s} \hat{t}) + R\dot{\theta} \hat{t} = -s\dot{\theta} \hat{r} \quad (\text{A4})$$

- (e) A2-rasmga murojaat qiling. Tezlik o'zgarishining $\Delta \vec{v}$ ($-\hat{t}$)-tashkil etuvchisi:

$$(-\hat{t}) \cdot \Delta \vec{v} = v \Delta \theta = v \dot{\theta} \Delta t.$$

Shuning uchun tezlanishning $\vec{a} = \Delta \vec{v} / \Delta t$ $\hat{t}$ -tashkil etuvchisi:

$$\hat{t} \cdot \vec{a} = -v \dot{\theta}.$$

(A4) tenglamaga ko'ra zarrachaning tezligi $v = s\dot{\theta}$ bo'lgani uchun, P nuqtadagi zarracha tezlanishining $\hat{t}$ -tashkil etuvchisi:

$$\vec{a} \cdot \hat{t} = -v \dot{\theta} = -(s\dot{\theta})\dot{\theta} = -s\dot{\theta}^2 \quad (\text{A5})$$

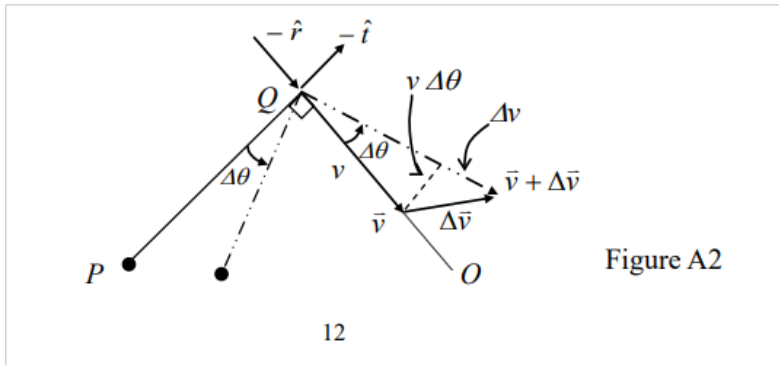


Figure 4: 1b-rasm (A2-rasm)

- (f) A3-rasmdan ko'rinib turibdiki, tezlanishning radial tashkil etuvchisi ham quyidagicha topilishi mumkin:

$$\vec{a} \cdot \hat{r} = -dv/dt = -d(s\dot{\theta})/dt.$$

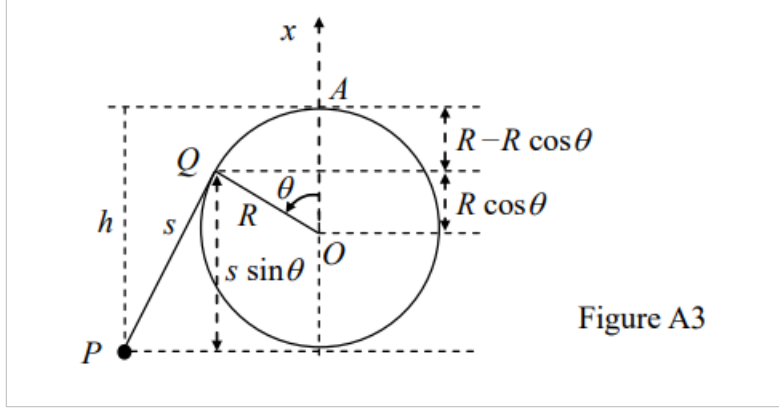


Figure 5: 1c-rasm (A3-rasm)

- (g) Zarrachaning tortishish potentsial energiyasi (gravitational potential energy) $U = -mgh$ formula bilan beriladi. Uni s va θ orqali ifodalash mumkin:

$$U(\theta) = -mg[R(1 - \cos \theta) + s \sin \theta] \quad (\text{A6})$$

- (h) Traektoriyaning eng quyi nuqtasida zarrachaning tortishish potentsial energiyasi U o'zining minimal qiymati U_m ga erishadi. (A6) tenglamani θ bo'yicha differensiallab va (A1) tenglamadan foydalanib, minimal energiyaga mos keluvchi θ_m burchak topiladi.

$$\begin{aligned} \frac{dU}{d\theta} &= -mg \left(R \sin \theta + \frac{ds}{d\theta} \sin \theta + s \cos \theta \right) \\ &= -mg[R \sin \theta + (-R) \sin \theta + s \cos \theta] \\ &= -mgs \cos \theta \end{aligned}$$

$\theta = \theta_m$ da $\frac{dU}{d\theta} \Big|_{\theta_m} = 0$ bo'lgani uchun $\theta_m = \frac{\pi}{2}$. Zarracha traektoriyasining eng quyi nuqtasi A4-rasmda ko'rsatilgan, bunda QP ip bo'lagining uzunligi $s = L - \pi R/2$.

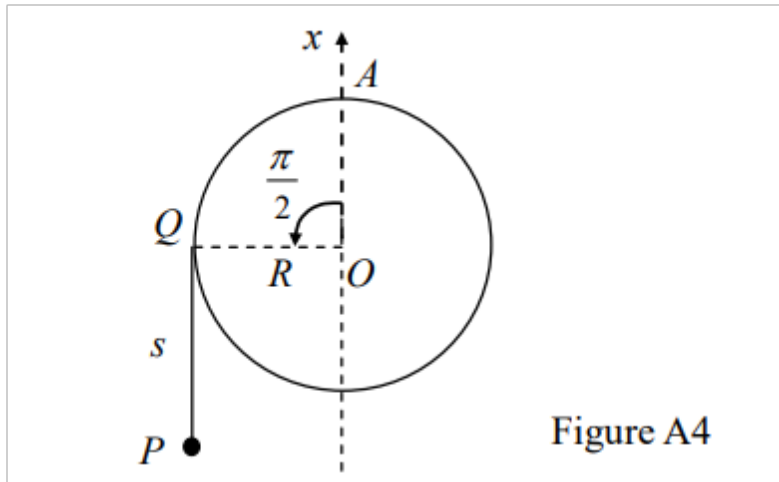


Figure 6: 1d-rasm (A4-rasm)

A4-rasm yoki (A6) tenglamadan minimal potentsial energiya:

$$U_m = U(\pi/2) = -mg[R + L - (\pi R/2)] \quad (A7)$$

Dastlabki to'liq mexanik energiya (total mechanical energy) $E = 0$. E saqlanganligi sababli, zarrachaning eng quyi nuqtadagi tezligi v_m quyidagi shartni qanoatlantiradi:

$$E = 0 = \frac{1}{2}mv_m^2 + U_m \quad (A8)$$

(A7) va (A8) tenglamalardan:

$$v_m = \sqrt{-2U_m/m} = \sqrt{2g[R + (L - \pi R/2)]} \quad (A9)$$

B qism

(h) (A6) tenglamadan zarrachaning to'liq mexanik energiyasini quyidagicha yozish mumkin:

$$E = 0 = \frac{1}{2}mv^2 + U(\theta) = \frac{1}{2}mv^2 - mg[R(1 - \cos \theta) + s \sin \theta] \quad (B1)$$

(A4) tenglamadan tezlik $v = s\dot{\theta}$ ga teng. Shuning uchun (B1) tenglama:

$$v^2 = (s\dot{\theta})^2 = 2g[R(1 - \cos \theta) + s \sin \theta] \quad (B2)$$

Ipning taranglik kuchi (tension) T bo'lsin. B1-rasmda ko'rsatilganidek, zarrachaga ta'sir etuvchi kuchlarning $\hat{t}$ -tashkil etuvchisi $-T + mg \sin \theta$ ga teng. (A5) tenglamaga ko'ra zarrachaning tangensial tezlanishi ($-s\ddot{\theta}^2$). Nyutonning ikkinchi qonunidan:

$$m(-s\ddot{\theta}^2) = -T + mg \sin \theta \quad (B3)$$

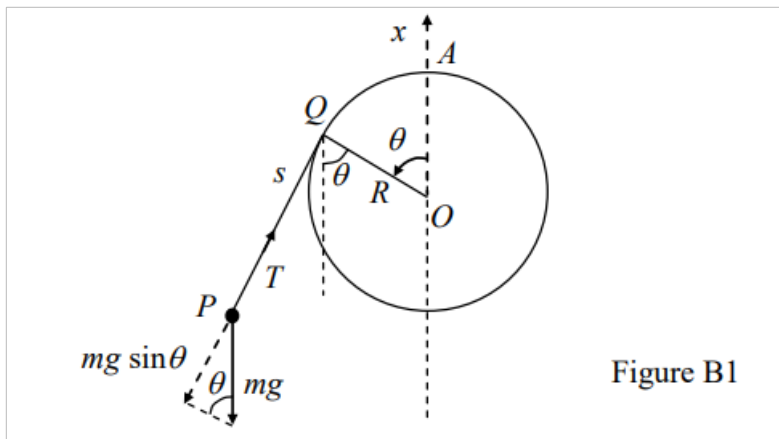


Figure 7: 2a-rasm (B1-rasm)

Oxirgi ikki tenglamaga asosan taranglik kuchi:

$$T = m(s\ddot{\theta}^2 + g \sin \theta) = \frac{mg}{s}[2R(1 - \cos \theta) + 3s \sin \theta]$$

$$\begin{aligned}
&= \frac{2mgR}{s} \left[\tan \frac{\theta}{2} - \frac{3}{2} \left(\theta - \frac{L}{R} \right) \right] \sin \theta \\
&= \frac{2mgR}{s} (y_1 - y_2) (\sin \theta)
\end{aligned} \tag{B4}$$

bu yerda $y_1 = \tan(\theta/2)$ va $y_2 = 3(\theta - L/R)/2$ funksiyalari B2-rasmda chizilgan.

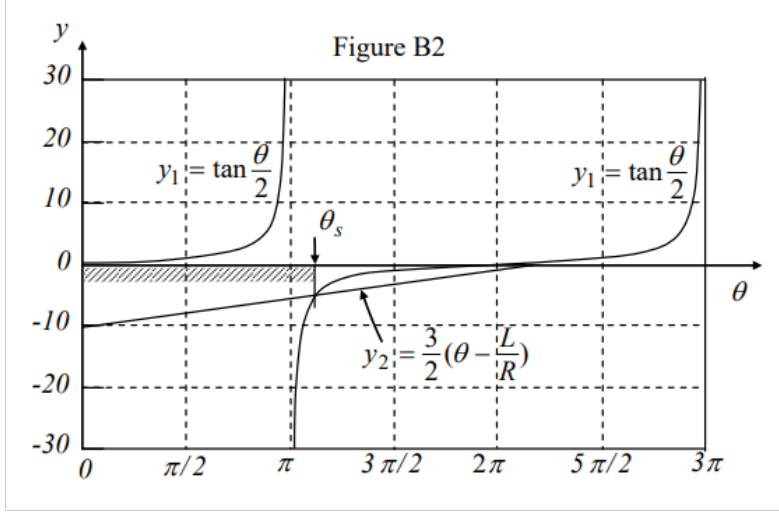


Figure 8: 2b-rasm (B2-rasm)

(B4) tenglama va B2-rasmdan B1-jadvalda keltirilgan natijalarni olamiz. $y_2 = y_1$ bo'ladigan burchak θ_s ($\pi < \theta_s < 2\pi$) deb ataladi va quyidagicha beriladi:

$$\frac{3}{2} \left(\theta_s - \frac{L}{R} \right) = \tan \frac{\theta_s}{2} \tag{B5}$$

yoki unga ekvivalent:

$$\frac{L}{R} = \theta_s - \frac{2}{3} \tan \frac{\theta_s}{2} \tag{B6}$$

L/R nisbati quyidagicha ma'lum:

$$\frac{L}{R} = \frac{9\pi}{8} + \frac{2}{3} \cot \frac{\pi}{16} = \left(\pi + \frac{\pi}{8} \right) - \frac{2}{3} \tan \frac{1}{2} \left(\pi + \frac{\pi}{8} \right) \tag{B7}$$

Oxirgi ikki tenglamadan $\theta_s = 9\pi/8$ ekanligi ko'rinadi.

Table 1: B1-jadval

	$(y_1 - y_2)$	$\sin \theta$	T taranglik
$0 < \theta < \pi$	musbat	musbat	musbat
$\theta = \pi$	$+\infty$	0	musbat
$\pi < \theta < \theta_s$	manfiy	manfiy	musbat
$\theta = \theta_s$	nol	manfiy	nol
$\theta_s < \theta < 2\pi$	musbat	manfiy	manfiy

B1-jadval shuni ko'rsatadiki, T taranglik kuchi $0 < \theta < \theta_s$ oraliqda musbat (ip tarang va to'g'ri). θ burchak θ_s ga yetganda $T = 0$ bo'ladi va sterjenga tegmay

turgan ip qismi keyin to'g'ri bo'lmay qoladi. QP kesma uzunligining eng kichik mumkin bo'lgan qiymati $s_{\min}$ shu $\theta = \theta_s$ da yuz beradi va quyidagicha topiladi:

$$s_{\min} = L - R\theta_s = R \left(\frac{9\pi}{8} + \frac{2}{3} \cot \frac{\pi}{16} - \frac{9\pi}{8} \right) = \frac{2R}{3} \cot \frac{\pi}{16} = 3.352R \quad (\text{B8})$$

$\theta = \theta_s$ bo'lganda $T = 0$ va (B2), (B3) tenglamalardan $v_s^2 = -g_{\min} \sin \theta_s$ kelib chiqadi. Shuning uchun tezlik v_s :

$$v_s = \sqrt{-g_{s\min} \sin \theta_s} = \sqrt{\frac{2gR}{3} \cot \frac{\pi}{16} \sin \frac{\pi}{8}} = \sqrt{\frac{4gR}{3} \cos \frac{\pi}{16}} \quad (\text{B9})$$

$$= 1.133\sqrt{gR}$$

- (h) $\theta \geq \theta_s$ bo'lganda zarracha og'irlik kuchi ta'sirida snaryad (projectile) kabi harakat qiladi. B3-rasmda ko'rsatilganidek, u boshlang'ich v_s tezlik bilan $P = (x_s, y_s)$ nuqtadan y -o'qi bilan $\phi = (3\pi/2 - \theta_s)$ burchak tashkil etgan yo'nalishda otiladi.

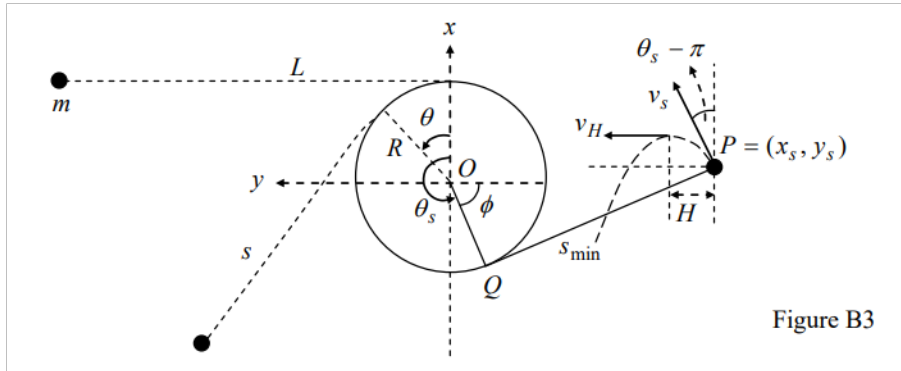


Figure 9: 2c-rasm (B3-rasm)

Zarrachaning parabolik traektoriyasining eng baland nuqtasidagi tezligi v_H uning otilishdagi boshlang'ich tezligining y -tashkil etuvchisiga teng:

$$v_H = v_s \sin(\theta_s - \pi) = \sqrt{\frac{4gR}{3}} \cos \frac{\pi}{16} \sin \frac{\pi}{8} = 0.4334\sqrt{gR} \quad (\text{B10})$$

P nuqtadan maksimal balandlik nuqtasigacha bo'lgan gorizonta masofa H :

$$H = \frac{v_s^2 \sin 2(\theta_s - \pi)}{2g} = \frac{v_s^2}{2g} \sin \frac{9\pi}{4} = 0.4535R \quad (\text{B11})$$

$\theta = \theta_s$ bo'lganda zarrachaning koordinatalari:

$$x_s = R \cos \theta_s - s_{\min} \sin \theta_s = -R \cos \frac{\pi}{8} + s_{\min} \sin \frac{\pi}{8} = 0.358R \quad (\text{B12})$$

$$y_s = R \sin \theta_s + s_{\min} \cos \theta_s = -R \sin \frac{\pi}{8} - s_{\min} \cos \frac{\pi}{8} = -3.478R \quad (\text{B13})$$

Ko‘rinib turibdiki, $|y_s| > (R+H)$. Shuning uchun zarracha sterjen sirtiga tegmay turib maksimal balandlikka erisha oladi.

C qism

(j) Faraz qilaylik, og'irlik dastlab O dan h masofa pastda joylashgan (C1-rasm).

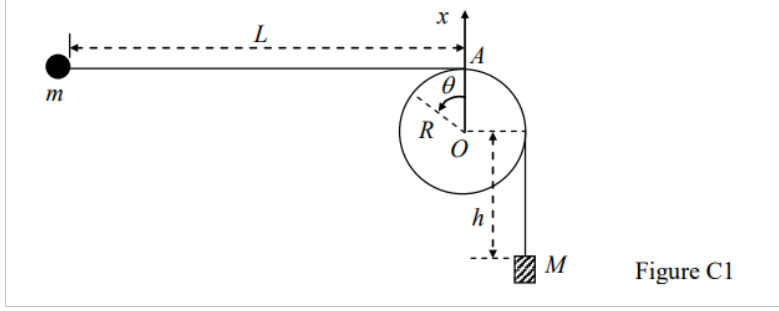


Figure 10: 3a-rasm (C1-rasm)

Og'irlik D masofaga tushib to'xtaganda, "zarracha+og'irlik" sistemasiga to'liq mexanik energiyaning saqlanish qonunini qo'llasak:

$$-Mgh = E' - Mg(h + D) \quad (C1)$$

bu yerda E' — og'irlik to'xtagandagi zarrachaning to'liq mexanik energiyasi. Bundan:

$$E' = MgD \quad (C2)$$

Ipning to'liq uzunligi Λ bo'lsin. Uning $\theta = 0$ dagi qiymati ixtiyoriy burchak θ dagi qiymatiga teng bo'lishi kerak:

$$\Lambda = L + \frac{\pi}{2}R + h = s + R\left(\theta + \frac{\pi}{2}\right) + (h + D) \quad (C3)$$

$D = \alpha L$ ekanligini va $\ell = L - D$ ni kiritib:

$$\ell = L - D = (1 - \alpha)L \quad (C4)$$

Oxirgi ikki tenglamadan:

$$s = L - D - R\theta = \ell - R\theta \quad (C5)$$

Og'irlik to'xtagandan keyin zarrachaning to'liq mexanik energiyasi saqlanadi. (C2) tenglamaga asosan, (B1) tenglama o'rniga endi:

$$E' = MgD = \frac{1}{2}mv^2 - mg[R(1 - \cos \theta) + s \sin \theta] \quad (C6)$$

Zarracha tezligining kvadrati:

$$v^2 = (s\dot{\theta})^2 = \frac{2MgD}{m} + 2gR\left[(1 - \cos \theta) + \frac{s}{R} \sin \theta\right] \quad (C7)$$

(B3) tenglama hali ham o'rinli bo'lgani uchun ipning taranglik kuchi:

$$-T + mg \sin \theta = m(-s\ddot{\theta}) \quad (C8)$$

Oxirgi ikki tenglamadan:

$$\begin{aligned}
T &= m(s\dot{\theta}^2 + g \sin \theta) \\
&= \frac{mg}{s} \left[\frac{2M}{m} D + 2R(1 - \cos \theta) + 3s \sin \theta \right] \\
&= \frac{2mgR}{s} \left[\frac{MD}{mR} + (1 - \cos \theta) + \frac{3}{2} \left(\frac{\ell}{R} - \theta \right) \sin \theta \right]
\end{aligned} \tag{C9}$$

(oxirgi tenglikda (C5) ishlatildi).

Endi quyidagi funktsiyani kiritamiz:

$$f(\theta) = 1 - \cos \theta + \frac{3}{2} \left(\frac{\ell}{R} - \theta \right) \sin \theta \tag{C10}$$

$\ell = (L - D) \gg R$ ekanligidan:

$$f(\theta) \approx 1 + \frac{3}{2} \frac{\ell}{R} \sin \theta - \cos \theta = 1 + A \sin(\theta - \phi) \tag{C11}$$

bu yerda:

$$A = \sqrt{1 + \left(\frac{3}{2} \frac{\ell}{R} \right)^2}, \quad \phi = \tan^{-1} \left(\frac{2R}{3\ell} \right) \tag{C12}$$

(C11) tenglamadan $f(\theta)$ ning minimal qiymati:

$$f_{\min} = 1 - A = 1 - \sqrt{1 + \left(\frac{3}{2} \frac{\ell}{R} \right)^2} \tag{C13}$$

Zarracha sterjen atrofida aylanayotganda taranglik kuchi T manfiy bo'lmashligi kerak. Shuning uchun (C9) tenglamadan:

$$\frac{MD}{mR} + f_{\min} = \frac{M(L - \ell)}{mR} + 1 - \sqrt{1 + \left(\frac{3\ell}{2R} \right)^2} \geq 0 \tag{C14}$$

yoki

$$\left(\frac{ML}{mR} \right) + 1 \geq \left(\frac{M\ell}{mR} \right) + \sqrt{1 + \left(\frac{3\ell}{2R} \right)^2} \approx \left(\frac{M\ell}{mR} \right) + \left(\frac{3\ell}{2R} \right) \tag{C15}$$

(C4) tenglamadan (C15) ni quyidagicha yozish mumkin:

$$\left(\frac{ML}{mR} \right) + 1 \geq \left(\frac{ML}{mR} + \frac{3L}{2R} \right) (1 - \alpha) \tag{C16}$$

(R/L) tartibidagi hadlarni e'tiborsiz qoldirib, bu tengsizlik:

$$\alpha \geq 1 - \frac{\left(\frac{ML}{mR} \right) + 1}{\left(\frac{ML}{mR} + \frac{3L}{2R} \right)} = \frac{\frac{3L}{2R} - 1}{\frac{ML}{mR} + \frac{3L}{2R}} = \frac{1 - \frac{2R}{3L}}{\frac{2M}{3m} + 1} \approx \frac{1}{1 + \frac{2M}{3m}} \tag{C17}$$

Shunday qilib, D/L nisbatining kritik qiymati:

$$\alpha_c = \frac{1}{1 + \frac{2M}{3m}} \tag{C18}$$

Baholash sxemasi (Marking Scheme)

Nazariy masala 1. Og'irlik tushayotgan belanchak

Table 2: Baholash jadvali

Umumiy ball	Qism ball	Javoblar uchun baholash mezonlari
6*4.3 ball	(a) 0.5	θ va $\dot{s}$ orasidagi munosabat ($\dot{s} = -R\dot{\theta}$). 0.2: $\dot{\theta} \propto \dot{s}$; 0.3: proporsionallik konstantasi ($-R$).
	(b) 0.7	Q ning O ga nisbatan tezligi ($\vec{v}_Q = R\dot{\theta}\hat{t}$). 0.2: modul $R\dot{\theta}$; 0.3: yo'nalish $\hat{t}$.
	(c) 0.7	Zarrachaning P nuqtadagi tezligi Q ga nisbatan ($\vec{v}' = -s\dot{\theta}\hat{r} + \dot{s}\hat{t}$). 0.2+0.1: $\hat{r}$ -tashkil etuvchining moduli va yo'nalishi; 0.3+0.1: $\hat{t}$ -tashkil etuvchining moduli va yo'nalishi.
	(d) 0.7	Zarrachaning P nuqtadagi tezligi O ga nisbatan ($\vec{v} = \vec{v}' + \vec{v}_Q = -s\dot{\theta}\hat{r}$). 0.3: $\vec{v}'$ va $\vec{v}_Q$ ni vektor qo'shish; 0.2+0.2: $\vec{v}$ ning moduli va yo'nalishi.
	(e) 0.7	Zarracha tezlanishining P nuqtadagi $\hat{t}$ -tashkil etuvchisi. 0.3: $\vec{a}$ yoki $\vec{a} \cdot \hat{t}$ ni tezlik bilan bog'lash, natijada $ \vec{a} \cdot \hat{t} = v^2/s$; 0.4: $\vec{a} \cdot \hat{t} = -s\dot{\theta}^2$ (0.1 minus ishorasi uchun).
	(f) 0.5	Potensial energiya U . 0.2: formula $U = -mgh$; 0.3: $h = R(1 - \cos \theta) + s \sin \theta$ yoki U ni θ, s, R orqali ifodalash.
6*4.3 ball	(g) 0.7	Eng quyi nuqtadagi tezlik v_m . 0.2: eng quyi nuqta $\theta = \pi/2$ yoki U minimal U_m ; 0.2: to'liq mexanik energiya $E = mv_m^2/2 + U_m = 0$; 0.3: $v_m = \sqrt{-2U_m/m} = \sqrt{2g[R + (L - \pi R/2)]}$.
	B qism	
	(h) 2.4	QP eng qisqa bo'lgandagi zarracha tezligi v_s . 0.4: QP eng qisqa bo'lganda $T = 0$; 0.3: harakat tenglamasi $-T + mg \sin \theta = m(-s\dot{\theta}^2)$; 0.3: $E = 0 = m(s\dot{\theta})^2/2 - mg[R(1 - \cos \theta) + s \sin \theta]$; 0.4: $\frac{3}{2}(\theta_s - L/R) = \tan(\theta_s/2)$; 0.5: $\theta_s = 9\pi/8$; 0.3+0.2: $v_s = \sqrt{4gR/3 \cos(\pi/16)} = 1.133\sqrt{gR}$.
6*3.4 ball	(i) 1.6	Zarrachaning eng baland nuqtadagi tezligi v_H . 0.4: $\theta \geq \theta_s$ da zarracha snaryad harakati; 0.3: otilish burchagi $\phi = (3\pi/2 - \theta_s)$; 0.3: v_H — tezlikning y -tashkil etuvchisi $\theta = \theta_s$ da; 0.4: zarracha sterjen sirtiga tegmasligini qayd etish; 0.3+0.2: $v_H = \sqrt{4gR/3 \cos(\pi/16)} \sin(\pi/8) = 0.4334\sqrt{gR}$.
	(j) 1.8	D/L nisbatining kritik qiymati α_c . 0.4: og'irlik to'xtaganda zarracha energiyasi $E' = MgD$; 0.3: $s = L - D - R\theta$; 0.3: $E' = MgD = mv^2/2 - mg[R(1 - \cos \theta) + s \sin \theta]$; 0.3: $-T + mg \sin \theta = m(-s\dot{\theta}^2)$; 0.3: T manfiy bo'lmasligi kerak degan xulosa; 0.6: D/L oraliq'ini aniqlovchi tengsizlik; 0.6: tengsizlikni yechib $\alpha = D/L$ oraliq'ini topish; 0.6: $\alpha_c = 1/(10 + 2M/(3m))$.

Nazariy masala 2

O'zgaruvchan kuchlanish ostidagi piezoelektrik kristall rezonator

Uzunligi (unstressed length) ℓ va ko'ndalang kesim yuzasi A bo'lgan bir jinsli sterjenni qaraymiz (2a-rasm). Uning uchlariga normal bo'ylab yo'nalgan, kattaligi teng va qarama-qarshi F kuchlari qo'yilganda uzunligi $\Delta\ell$ ga o'zgaradi. Uchlaridagi kuchlanish (stress) $T = F/A$ deb aniqlanadi. Uzunlikning nisbiy o'zgarishi (strain) $S = \Delta\ell/\ell$. Kuchlanish va deformatsiya orqali Guk qonuni (Hooke's law) quyidagicha ifodalanadi:

$$T = YS \quad \text{yoki} \quad \frac{F}{A} = Y \frac{\Delta\ell}{\ell} \quad (1)$$

bu yerda Y — sterjen materialining Young moduli (Young's modulus). Siqiluvchi kuchlanish (compressive stress) $F < 0$ va uzunlikning kamayishi ($\Delta\ell < 0$) ga to'g'ri keladi. Bunday kuchlanish manfiy qiymatga ega va bosim p bilan $T = -p$ munosabat orqali bog'langan.

Zichligi ρ bo'lgan bir jinsli sterjen uchun bo'ylama to'lqinlarning (longitudinal waves) tarqalish tezligi (tovush tezligi) quyidagicha:

$$u = \sqrt{Y/\rho} \quad (2)$$

So'nish va dissipatsiya (damping and dissipation) ta'sirini e'tiborsiz qoldirish mumkin.

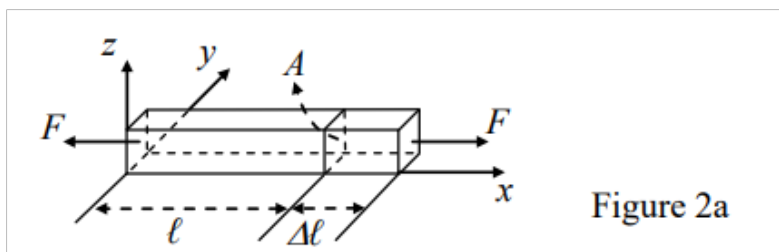


Figure 11: 2a-rasm

A qism: mexanik xususiyatlar

Yarim cheksiz uzunlikdagi (semi-infinite) bir jinsli sterjen $x = 0$ dan ∞ gacha cho'zilgan (2b-rasm). Uning zichligi ρ . Dastlab sterjen qo'zg'almas va kuchlanishsiz (unstressed). Keyin piston uning $x = 0$ nuqtasidagi chap tomoniga juda qisqa vaqt Δt davomida kichik bosim p ni ta'sir ettiradi, natijada bosim to'lqini o'ng tomonga u tezlik bilan tarqaladi.

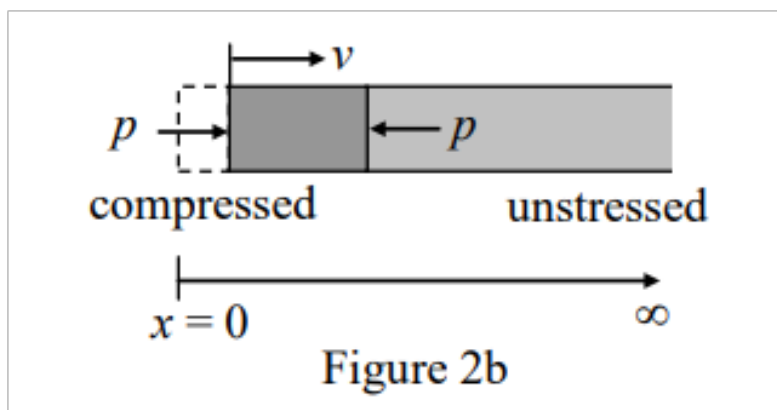


Figure 12: 2b-rasm

- (a) Agar piston sterjenning chap tomonini o'zgaras tezlik v bilan harakatlantirsa (2b-rasm), Δt vaqt davomida chap tomondagi deformatsiya (strain) S va bosim p nimaga teng? Javoblar faqat ρ , u va v orqali ifodalansin. (1.6 ball)
- (b) x yo'nalishi bo'ylab tarqalayotgan bo'ylama to'lqinni qaraylik. Sterjen kuchlanishsiz holatda bo'lgandagi (2c-rasm) x nuqtadagi ko'ndalang kesimning t vaqtdagi siljishi $\xi(x, t)$ bo'lsin va

$$\xi(x, t) = \xi_0 \sin k(x - ut) \quad (3)$$

deb olaylik, bu yerda ξ_0 va k — o'zgaraslar. Tegishli tezlik $v(x, t)$, deformatsiya $S(x, t)$ va bosim $p(x, t)$ ni x va t funksiyasi sifatida aniqlang. (2.4 ball)

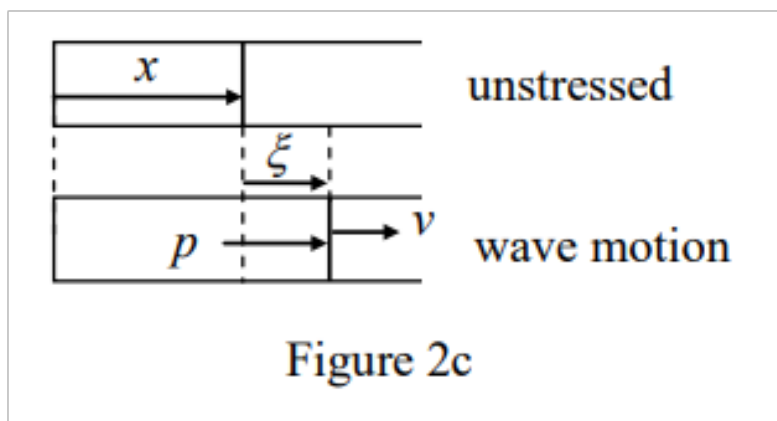


Figure 13: 2c-rasm

B qism: elektromexanik xususiyatlar (piezoelektrik effekt bilan birga)

Uzunligi b , qalinligi h va eni w bo'lgan kvarts kristalli plastinkani qaraymiz (2d-rasm). Uning uzunligi va qalinligi mos ravishda x -o'qi va z -o'qi bo'ylab joylashgan. Elektrod-lar yuqori va pastki sirlarga yupqa metall qoplamalar sifatida yasalgan. Elektrod-larning markazlariga (ular x yo'nalishidagi bo'ylama tebranishlar uchun qo'zg'almas deb hisoblanadi) lehimlangan elektr simlari ham mahkamlash vazifasini bajaradi (2e-rasm).

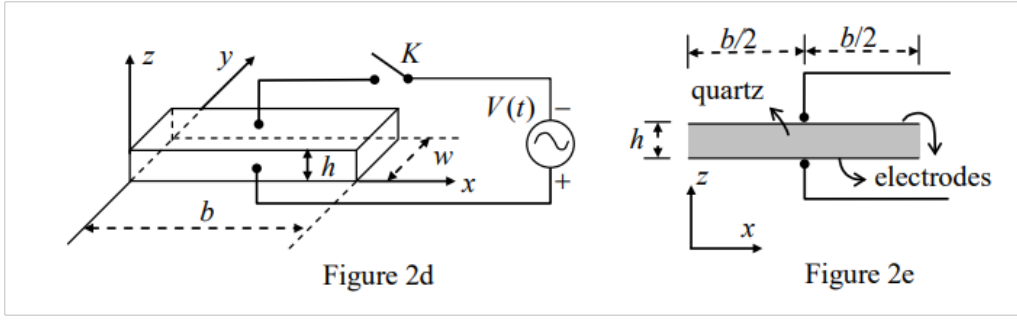


Figure 14: 2d-rasm, 2e-rasm

Ko'rilayotgan kvarts kristallining zichligi $\rho = 2.65 \times 10^3 \text{ kg/m}^3$ va Young moduli $Y = 7.87 \times 10^{10} \text{ N/m}^2$. Plastinkaning uzunligi $b = 1.00 \text{ cm}$, eni w va balandligi h shundayki, $h \ll w$ va $w \ll b$. Kalit K ochiq holda, biz faqat kvarts plastinkasida x yo'nalishidagi bo'ylama turgan to'lqin (standing wave) rejimlari qo'zg'atiladi deb faraz qilamiz.

Chastotasi $f = \omega/2\pi$ bo'lgan turgan to'lqin uchun muvozanat holati x bo'lgan ko'ndalang kesimning t vaqtdagi siljishi $\xi(x, t)$ quyidagicha yozilishi mumkin:

$$\xi(x, t) = 2\xi_0 g(x) \cos \omega t, \quad (0 \leq x \leq b) \quad (4a)$$

bu yerda ξ_0 musbat o'zgarmas va fazoviy funksiya $g(x)$ quyidagi ko'rinishga ega:

$$g(x) = B_1 \sin k \left(x - \frac{b}{2} \right) + B_2 \cos k \left(x - \frac{b}{2} \right) \quad (4b)$$

$g(x)$ ning maksimal qiymati 1 ga teng va $k = \omega/u$. Yodda tuting: elektrodning markazlari qo'zg'almas, plastinkaning chap va o'ng tomonlari erkin va ularda kuchlanish (bosim) nolga teng.

- (c) Kvarts plastinkasidagi bo'ylama turgan to'lqin uchun (4b) tenglamadagi B_1 va B_2 qiymatlarini aniqlang. (1.2 ball)
- (c) Kvarts plastinkasida bo'ylama turgan to'lqinlar qo'zg'alishi mumkin bo'lgan eng past ikki chastotani toping. (1.2 ball)

Piezoelektrik effekt — kvarts kristallining maxsus xususiyati. Kristallning siqilishi yoki kengayishi uning bo'ylab elektr kuchlanishini hosil qiladi va aksincha, kristallga tashqi kuchlanish berilsa, kuchlanish qutbliligiga qarab kristall kengayadi yoki siqiladi. Shunday qilib, mexanik va elektr tebranishlar bir-biriga bog'lanib, kvarts kristalli orqali rezonansga kelishi mumkin.

Piezoelektrik effektini hisobga olish uchun, yuqori va pastki elektrodlardagi sirt zaryad zichliklari mos ravishda $-\sigma$ va $+\sigma$ bo'lsin, bunda kvarts plastinkasi z yo'nalishida E elektr maydoni ostida. Plastinkaning x yo'nalishidagi deformatsiyasi va kuchlanishini mos ravishda S va T bilan belgilaymiz. U holda kvarts kristallining piezoelektrik effekti quyidagi tenglamalar bilan tavsiflanadi:

$$S = (1/Y)T + d_p E \quad (5a)$$

$$\sigma = d_p T + \epsilon_p E \quad (5b)$$

bu yerda $1/Y = 1.27 \times 10^{-11} \text{ m}^2/\text{N}$ — o'zgaras elektr maydonidagi elastik moslik (Young modulining teskarisi), $\epsilon_p = 4.06 \times 10^{-11} \text{ F/m}$ — o'zgaras kuchlanishdagi dielektrik o'tkazuvchanlik (permittivity), $d_p = 2.25 \times 10^{-12} \text{ m/V}$ — piezoelektrik ko'effitsient.

2d-rasmdagi kalit K ni yopamiz. O'zgaruvchan kuchlanish $V(t) = V_m \cos \omega t$ endi elektrodarga beriladi va kvarts plastinkasida z yo'nalishida bir jinsli elektr maydoni $E(t) = V(t)/h$ hosil bo'ladi. Qo'zg'almas holat (steady state) ga erishilganda, plastinkada x yo'nalishida burchak chastotasi ω bo'lgan bo'ylama turgan to'lqin paydo bo'ladi.

E bir jinsli bo'lganda, bo'ylama turgan to'lqinning to'lqin uzunligi λ va chastotasi f hali ham $\lambda = u/f$ munosabat bilan bog'liq, bu yerda u (2) tenglama bilan berilgan. Ammo (5a) tenglama ko'rsatganidek, $T = YS$ endi o'rinli emas, garchi deformatsiya va kuchlanish ta'riflari o'zgarishsiz qolsa va plastinkaning uch tomonlari erkin (kuchlanish nol) bo'lsa.

- (e) (5a) va (5b) tenglamalarni hisobga olib, pastki elektrodagi sirt zaryad zichligi $\sigma(x, t)$ quyidagi ko'rinishga ega:

$$\sigma(x, t) = \left[D_1 \cos k \left(x - \frac{b}{2} \right) + D_2 \right] \frac{V(t)}{h}$$

bu yerda $k = \omega/u$. D_1 va D_2 ifodalarini toping. (2.2 ball)

- (e) Pastki elektrodagi to'liq sirt zaryadi $Q(t)$ bilan $V(t)$ orasidagi bog'lanish:

$$Q(t) = \left[1 + \alpha^2 \left(\frac{2}{kb} \tan \frac{kb}{2} - 1 \right) \right] C_0 V(t) \quad (6)$$

C_0 ifodasini hamda α^2 ifodasi va uning son qiymatini toping. (1.4 ball)

[Javoblar Varaqasi]

2-Nazariy Savol

O'zgaruvchan kuchlanish ostidagi pyezoelektrik kristall rezonator

So'ralgan joyda, har bir javobni analytical expressions (analitik ifodalar) ko'rinishida yozing va undan keyin numerical values and units (sonli qiymatlar hamda birliklar)ni keltiring. Misol uchun: doira yuzi $A = \pi r^2 = 1.23 \text{ m}^2$.

- (a) Chap tomondagi S strain (nisbiy deformatsiya) va p pressure (bosim) quyidagilarga teng (ρ , u , va v orqali ifodalang):

$S =$

$p =$

(b) $v(x, t)$ velocity (tezlik), $S(x, t)$ strain (nisbiy deformatsiya) va $p(x, t)$ pressure (bosim) quyidagilarga teng:

$v(x, t) =$
$S(x, t) =$
$p(x, t) =$

(c) B_1 va B_2 ning qiymatlari quyidagilarga teng:

$B_1 =$
$B_2 =$

(d) Turg'un to'lqinlarning eng kichik ikkita frequency (chastotasi) quyidagilarga teng (ifodasi va qiymati):

Eng past (The Lowest):
Ikkinchi eng past (The Second Lowest):

(e) D_1 va D_2 ning ifodalari quyidagilarga teng:

(f) α^2 o'zgarmasi (ifodasi va qiymati) hamda C_0 o'zgarmasi (faqat ifodasi) quyidagilarga teng:

$$D_1 =$$

$$D_2 =$$

$$\alpha^2 =$$

$$C_0 =$$

2-masala yechimi

A qism

- (a) 2a-rasmga murojaat qiling. Sterjenning chap tomoni $v\Delta t$ masofaga siljiydi, bosim to'liqini esa $u\Delta t$ masofaga tarqaladi, bunda $u = \sqrt{Y/\rho}$. Chap tomondagi deformatsiya:

$$S = \frac{\Delta \ell}{\ell} = \frac{-v\Delta t}{u\Delta t} = -\frac{v}{u} \quad (\text{A1a})$$

Guk qonunidan chap tomondagi bosim:

$$p = -YS = Y\frac{v}{u} = \rho uv \quad (\text{A1b})$$

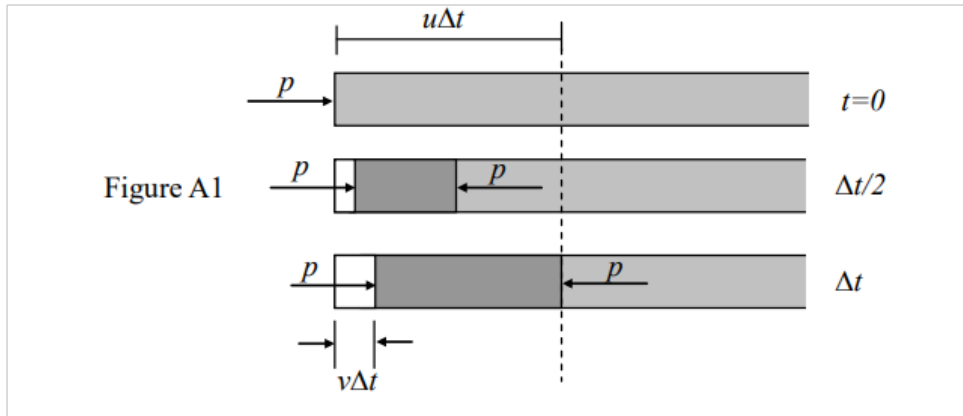


Figure 15: 2a-rasm (yechim)

- (b) Tezlik v siljish ξ bilan oddiy garmonik harakat (yoki 2b-rasmga ko'rsatilgan tekis aylanma harakat) kabi bog'langan, burchak chastotasi $\omega = ku$. Shuning uchun, agar $\xi(x, t) = \xi_0 \sin k(x - ut)$ bo'lsa, u holda

$$v(x, t) = -ku\xi_0 \cos k(x - ut) \quad (\text{A2})$$

Deformatsiya va bosim tezlik bilan (a) masaladagi kabi bog'langan. Demak,

$$S(x, t) = -v(x, t)/u = k\xi_0 \cos k(x - ut) \quad (\text{A3})$$

$$p(x, t) = \rho uv(x, t) = -k\rho u^2 \xi_0 \cos k(x - ut) = -YS(x, t) = -kY \xi_0 \cos k(x - ut) \quad (\text{A4})$$

Alternativ ravishda, javoblarni differentsiallashtirish orqali ham olish mumkin:

$$v(x, t) = \frac{\partial \xi}{\partial t} = -ku \xi_0 \cos k(x - ut), \quad S(x, t) = \frac{\partial \xi}{\partial x} = k \xi_0 \cos k(x - ut), \quad p(x, t) = -Y \frac{\Delta \xi}{\Delta x} = -kY \xi_0 \cos k(x - ut)$$

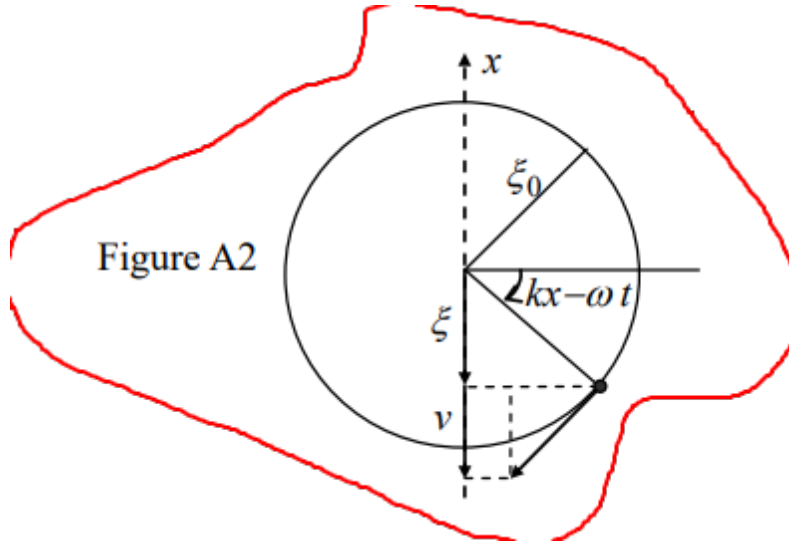


Figure 16: 2b-rasm (yechim)

B qism

- (c) Burchak chastotasi ω va tarqalish tezligi u berilgan, to'liq uzunligi $\lambda = 2\pi/k$ bilan $k = \omega/u$. Siljishning fazoviy o'zgarishi ξ quyidagicha:

$$g(x) = B_1 \sin k \left(x - \frac{b}{2} \right) + B_2 \cos k \left(x - \frac{b}{2} \right)$$

Elektrodlarning markazlari qo'zg'almas deb hisoblangani uchun $g(b/2) = 0$. Bundan $B_2 = 0$ kelib chiqadi. $g(x)$ ning maksimal qiymati 1 ga teng bo'lgani uchun $B_1 = \pm 1$ va

$$g(x) = \pm \sin \frac{\omega}{u} \left(x - \frac{b}{2} \right)$$

Shunday qilib, siljish:

$$\xi(x, t) = \pm 2\xi_0 \sin \frac{\omega}{u} \left(x - \frac{b}{2} \right) \cos \omega t$$

- (c) Bosim (yoki kuchlanish) kvarts plastinkasining uch tomonlarida ($x = 0$ va $x = b$) nolga teng bo'lishi kerak. Shuning uchun bu masalaning javobini uzunligi b bo'lgan ochiq naydagi tovush to'liqlarining rezonans chastotalariga o'xshab olish mumkin. Ammo elektrodlarning markazlari qo'zg'almas bo'lgani uchun, asosiy tonning barcha juft garmonikalari (even harmonics) chiqarib tashlanadi, chunki

ular plastinkaning teng ikkiga bo'luvchi tekisligida siljishning tugun (node) emas, balki do'nglik (antinode) nuqtasiga ega.

Asosiy ton to'liq uzunligi $\lambda = 2b$ ga teng, shuning uchun asosiy chastota:

$$f_1 = \frac{u}{2b}$$

Tarqalish tezligi:

$$u = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{7.87 \times 10^{10}}{2.65 \times 10^3}} = 5.45 \times 10^3 \text{ m/s}$$

$b = 1.00 \times 10^{-2} \text{ m}$ berilgan, eng past ikki turgan to'liq chastotasi:

$$f_1 = \frac{u}{2b} = \frac{5.45 \times 10^3}{2 \times 1.00 \times 10^{-2}} = 2.73 \times 10^5 \text{ Hz} = 273 \text{ kHz}$$

$$f_3 = 3f_1 = \frac{3u}{2b} = 8.18 \times 10^5 \text{ Hz} = 818 \text{ kHz}$$

(Eslatma: faqat toq garmonikalar $n = 1, 3, 5, \dots$ mumkin.)

Alternativ yechim: Bo'ylama turgan to'liq $x = b/2$ da siljish tuguniga ega. Uni qarama-qarshi yo'nalishda tarqalayotgan ikki to'liqning yig'indisi sifatida qarash mumkin. Natijada $\xi(x, t) = 2\xi_m \sin k(x - b/2) \cos \omega t$ ko'rinishi kelib chiqadi. Uchlarda kuchlanish nol bo'lish sharti $\cos(kb/2) = 0$ ni beradi, bundan $kb = n\pi, n = 1, 3, 5, \dots$ va chastota $f = nu/(2b)$.

(c) va (d) masalalarning muqobil yechimi — Yee Lin Chiu

Kvarts plitasidagi bo'ylama turgan to'liq $x = b/2$ da siljish tuguniga ega. Uni qarama-qarshi yo'nalishlarda tarqalayotgan ikki to'liqning yig'indisi deb qarash mumkin. Shuning uchun uning siljishi va tezligi quyidagi ko'rinishga ega bo'lishi kerak:

$$\begin{aligned} \xi(x, t) &= \xi_m \left[\sin k \left(x - \frac{b}{2} - ut \right) + \sin k \left(x - \frac{b}{2} + ut \right) \right] \\ &= 2\xi_m \sin k \left(x - \frac{b}{2} \right) \cos \omega t \\ v(x, t) &= -ku\xi_m \left[\cos k \left(x - \frac{b}{2} - ut \right) - \cos k \left(x - \frac{b}{2} + ut \right) \right] \\ &= -2\omega\xi_m \sin k \left(x - \frac{b}{2} \right) \sin \omega t \end{aligned}$$

bu yerda $\omega = ku$ va kvadrat qavs ichidagi birinchi va ikkinchi ko'paytuvchilar mos ravishda $+x$ va $-x$ yo'nalishlarida tarqalayotgan to'liqlarni ifodalaydi. Eslatib o'tamiz, agar $\xi_m = \pm\xi_0$ deb olsak, (B6) tenglama (B3) tenglama bilan bir xil bo'ladi.

$-x$ yo'nalishida tarqalayotgan to'lqin uchun (A1a) va (A1b) tenglamalarda v o'rniga $-v$ qo'yilishi kerak, shuning uchun:

$$S = -\frac{v}{u} \quad \text{va} \quad p = \rho uv \quad (+x \text{ yo'nalishida tarqalayotgan to'lqin})$$

$$S = \frac{v}{u} \quad \text{va} \quad p = -\rho uv \quad (-x \text{ yo'nalishida tarqalayotgan to'lqin})$$

(b) masaladagidek, deformatsiya va bosim quyidagicha ifodalanadi:

$$\begin{aligned} S(x, t) &= -k\xi_m \left[-\cos k \left(x - \frac{b}{2} - ut \right) - \cos k \left(x - \frac{b}{2} + ut \right) \right] \\ &= 2k\xi_m \cos k \left(x - \frac{b}{2} \right) \cos \omega t \end{aligned}$$

$$\begin{aligned} p(x, t) &= -\rho u \omega \xi_m \left[\cos k \left(x - \frac{b}{2} - ut \right) + \cos k \left(x - \frac{b}{2} + ut \right) \right] \\ &= -2\rho u \omega \xi_m \cos k \left(x - \frac{b}{2} \right) \cos \omega t \end{aligned}$$

Eslatib o'tamiz, v , S , va p ni (b) masaladagidek ξ ni differensiallab ham olish mumkin.

Plitaning har ikki uchida ($x = 0$ va $x = b$) kuchlanish T yoki bosim p vaqtning barcha momentlarida nolga teng bo'lishi kerak, chunki ular erkindir. (B11) tenglamadan bu faqat $\cos(kb/2) = 0$ bo'lganda mumkin, ya'ni

$$kb = \frac{\omega}{u}b = \frac{2\pi f}{\lambda}b = n\pi, \quad n = 1, 3, 5, \dots$$

To'lqin uzunligi λ orqali (B12) tenglama quyidagicha yozilishi mumkin:

$$\lambda = \frac{2b}{n}, \quad n = 1, 3, 5, \dots$$

Chastota esa quyidagicha ifodalanadi:

$$f = \frac{u}{\lambda} = \frac{nu}{2b} = \frac{n}{2b} \sqrt{\frac{\nu}{\rho}}, \quad n = 1, 3, 5, \dots$$

Bu (B4) va (B5) tenglamalarda berilgan natijalar bilan bir xildir.

- (c) (e) Masala savolidagi (5a) va (5b) tenglamalardan piezoelektrik effekt quyidagi tenglamalarga olib keladi:

$$T = Y(S - d_p E) \quad (B15)$$

$$\sigma = Y d_p S + \epsilon_T \left(1 - Y \frac{d_p^2}{\epsilon_T} \right) E \quad (B16)$$

$x = b/2$ har qanday bo'ylama turgan to'lqin uchun siljish tuguni bo'lgani uchun, siljish ξ va deformatsiya S quyidagi ko'rinishga ega (B6 va B10 tenglamalardan, $\omega = ku$):

$$\xi(x, t) = \xi_m \sin k \left(x - \frac{b}{2} \right) \cos(\omega t + \phi) \quad (B17)$$

$$S(x, t) = k \xi_m \cos k \left(x - \frac{b}{2} \right) \cos(\omega t + \phi) \quad (B18)$$

bu yerda vaqt funksiyalariga faza o'zgarmasi ϕ qo'shilgan.

Farazga ko'ra, elektrodlar orasidagi elektr maydoni E bir jinsli va faqat vaqtga bog'liq:

$$E(x, t) = \frac{V(t)}{h} = \frac{V_m \cos \omega t}{h} \quad (B19)$$

(B18) va (B19) ni (B15) ga qo'yib:

$$T = Y \left[k \xi_m \cos k \left(x - \frac{b}{2} \right) \cos(\omega t + \phi) - \frac{d_p}{h} V_m \cos \omega t \right] \quad (B20)$$

Kuchlanish T plastinkaning ikkala uchida ($x = 0$ va $x = b$) hamma vaqt nolga teng bo'lishi kerak (chunki ular erkin). Bu faqat $\phi = 0$ va

$$k \xi_m \cos \frac{kb}{2} = d_p \frac{V_m}{h} \quad (B21)$$

bo'lganda mumkin.

$\phi = 0$ bo'lgani uchun (B16), (B18) va (B19) dan sirt zaryad zichligi vaqtga bog'liqligi $\cos \omega t$ bilan bir xil ekanligi kelib chiqadi va uni quyidagicha ifodalash mumkin:

$$\sigma(x, t) = \sigma(x) \cos \omega t \quad (B22)$$

bunda x ga bog'liqlik:

$$\begin{aligned} \sigma(x) &= Y d_p k \xi_m \cos k \left(x - \frac{b}{2} \right) + \epsilon_T \left(1 - Y \frac{d_p^2}{\epsilon_T} \right) \frac{V_m}{h} \\ &= \left[Y \frac{d_p^2}{h} \frac{V_m}{h} ? \text{ To'g'rilash: } \xi_m \text{ ni (B21) dan topsak: } k \xi_m = \frac{d_p V_m}{h \cos(kb/2)}. \text{ Shuning uchun} \right. \\ \sigma(x) &= Y d_p \cdot \frac{d_p V_m}{h \cos(kb/2)} \cos k \left(x - \frac{b}{2} \right) + \epsilon_T \left(1 - Y \frac{d_p^2}{\epsilon_T} \right) \frac{V_m}{h} \\ &= \frac{V_m}{h} \left[Y d_p^2 \frac{\cos k(x - b/2)}{\cos(kb/2)} + \epsilon_T \left(1 - Y \frac{d_p^2}{\epsilon_T} \right) \right] \end{aligned}$$

Shuning uchun (e) qismda berilgan shaklga mos keladigan:

$D_1 = Yd_p^2 / \cos(kb/2)$? Lekin ular $D_1 \cos k(x-b/2) + D_2$ ko'rinishida so'ragan. Demak

$$D_1 = \frac{Yd_p^2}{\cos(kb/2)}, \quad D_2 = \epsilon_T - Yd_p^2 = \epsilon_T(1 - \alpha^2)$$

Bu yerda $\alpha^2 = Yd_p^2 / \epsilon_T$ (keyingi qismda aniqlanadi). (f) qismdagi natijaga mos kelish uchun, ular (6) tenglamada $\alpha^2(2/(kb) \tan(kb/2) - 1) + 1$ ko'rinishini olgan. Bizning ifodamizni integrallab shu ko'rinishga kelish mumkin.

(c) To'liq sirt zaryadi $Q(t)$ ni pastki elektrod yuzasi bo'yicha $\sigma(x, t)$ ni integrallab topamiz:

$$Q(t) = \int_0^b \sigma(x, t) w dx = \frac{w}{h} V_m \cos \omega t \int_0^b \left[\frac{Yd_p^2}{\cos(kb/2)} \cos k \left(x - \frac{b}{2} \right) + \epsilon_T(1 - \alpha^2) \right] dx$$

Integral:

$$\int_0^b \cos k \left(x - \frac{b}{2} \right) dx = \frac{2}{k} \sin \frac{kb}{2}$$

Demak,

$$\frac{Q(t)}{V(t)} = \frac{w}{h} \left[\frac{Yd_p^2}{\cos(kb/2)} \cdot \frac{2}{k} \sin \frac{kb}{2} + \epsilon_T(1 - \alpha^2)b \right] = \frac{w}{h} \left[Yd_p^2 \frac{2}{k} \tan \frac{kb}{2} + \epsilon_T b(1 - \alpha^2) \right]$$

$\alpha^2 = Yd_p^2 / \epsilon_T$ ni qo'llab:

$$\frac{Q}{V} = \frac{w}{h} \epsilon_T b \left[\alpha^2 \frac{2}{kb} \tan \frac{kb}{2} + (1 - \alpha^2) \right] = C_0 \left[1 + \alpha^2 \left(\frac{2}{kb} \tan \frac{kb}{2} - 1 \right) \right]$$

bu yerda

$$C_0 = \epsilon_T \frac{bw}{h}, \quad \alpha^2 = Y \frac{d_p^2}{\epsilon_T}$$

Son qiymatlarni qo'yib:

$$\alpha^2 = \frac{(7.87 \times 10^{10}) \cdot (2.25 \times 10^{-12})^2}{4.06 \times 10^{-11}} = \frac{7.87 \times 10^{10} \cdot 5.0625 \times 10^{-24}}{4.06 \times 10^{-11}} = \frac{3.984 \times 10^{-13}}{4.06 \times 10^{-11}} = 9.81 \times 10^{-3}$$

$$\alpha^2 \approx 9.82 \times 10^{-3}$$

C_0 ifodasi: $C_0 = \epsilon_T \frac{bw}{h}$. (Bu statik limit $k \rightarrow 0$ da $Q/V \rightarrow C_0$ ekanligidan ham ko'rinadi.)

Javoblar varaqasi (2-masala)

$$(a) \quad S = -\frac{v}{u}, \quad p = \rho uv$$

$$(b) \quad v(x, t) = -ku\xi_0 \cos k(x - ut), \quad S(x, t) = k\xi_0 \cos k(x - ut), \quad p(x, t) = -kY\xi_0 \cos k(x - ut)$$

$$(c) \quad B_1 = \pm 1, \quad B_2 = 0$$

$$(d) \quad \text{Eng past: } f_1 = \frac{u}{2b} = 273 \text{ kHz; Ikkinchi eng past: } f_3 = \frac{3u}{2b} = 818 \text{ kHz}$$

$$(e) \quad D_1 = \frac{Yd_p^2}{\cos(kb/2)}, \quad D_2 = \epsilon_T(1 - \alpha^2)$$

$$(f) \quad \alpha^2 = Y \frac{d_p^2}{\epsilon_T} = 9.82 \times 10^{-3}, \quad C_0 = \epsilon_T \frac{bw}{h}$$

Nazariy masala 3

Neytrino massasi va neytron yemirilishi

Erkin neytron massasi m_n bo'lib, laboratoriya sanoq tizimida tinch holatda (at rest) uchta o'zaro ta'sirlashmaydigan zarrachaga: protonga, elektronga va anti-neytrinoga yemiriladi. Protonning tinch massasi m_p , anti-neytrinoning tinch massasi $m_{\bar{\nu}}$ noldan farqli va elektronning tinch massasi m_e dan ancha kichik deb hisoblanadi. Vakuumdagi yorug'lik tezligini c bilan belgilaymiz. Massalarning o'lchangan qiymatlari:

$$m_n = 939.56563 \text{ MeV}/c^2, \quad m_p = 938.27231 \text{ MeV}/c^2, \quad m_e = 0.5109907 \text{ MeV}/c^2$$

Quyida barcha energiyalar va tezliklar laboratoriya sanoq tizimiga oid. Yemirilish natijasida chiqqan elektronning to'liq energiyasi E bo'lsin.

- (a) E ning mumkin bo'lgan maksimal qiymati $E_{\max}$ ni va $E = E_{\max}$ bo'lgandagi anti-neytrinoning tezligi v_m ni toping. Ikkala javob ham zarrachalarning tinch massalari va yorug'lik tezligi orqali ifodalansin. $m_{\bar{\nu}} < 7.3 \text{ eV}/c^2$ ekanligidan foydalanib, $E_{\max}$ va v_m/c nisbatini 3 ta muhim raqamgacha hisoblang. (4.0 ball)

3-masala yechimi

Neytron tinch holatda yemiriladi. To'liq energiya va impulsning saqlanish qonunlaridan foydalanamiz. Neytronning dastlabki to'liq energiyasi $m_n c^2$, dastlabki impulsi nol.

Yemirilish mahsulotlari: proton (p), elektron (e) va anti-neytrino ($\bar{\nu}$). Ularning energiyalari: $E_p, E_e, E_{\bar{\nu}}$; impulsarlari: $\vec{p}_p, \vec{p}_e, \vec{p}_{\bar{\nu}}$.

Saqlanish qonunlari:

$$E_p + E_e + E_{\bar{\nu}} = m_n c^2 \quad (1)$$

$$\vec{p}_p + \vec{p}_e + \vec{p}_{\bar{\nu}} = 0 \quad (2)$$

Har bir zarracha uchun relativistik energiya-impuls munosabati:

$$E_p^2 = (m_p c^2)^2 + (p_p c)^2, \quad E_e^2 = (m_e c^2)^2 + (p_e c)^2, \quad E_{\bar{\nu}}^2 = (m_{\bar{\nu}} c^2)^2 + (p_{\bar{\nu}} c)^2$$

Elektron energiyasining maksimal qiymati anti-neytrino va proton bir-biriga qarama-qarshi yo'nalishda va impulsning kattaligi eng kichik bo'lganda yuz beradi. Maksimal elektron energiyasida anti-neytrino va protonning nisbiy impulsi minimal bo'ladi. Bu holda proton va anti-neytrino bir xil yo'nalishda harakatlanadi (impulsning vektor yig'indisi nol bo'lishi uchun elektron qarama-qarshi tomonga ketadi). Ikki zarrachaning (proton va anti-neytrino) birgalikdagi massasi minimal bo'lganda, elektron maksimal energiyaga ega bo'ladi. Ya'ni proton va anti-neytrino tinch holatiga nisbatan bir-biriga nisbatan tinch holatda bo'lganda (ularning nisbiy impulsi nol), u holda ularning umumiy energiyasi ularning tinch massalari yig'indisiga teng bo'ladi: $E_p + E_{\bar{\nu}} = (m_p + m_{\bar{\nu}})c^2$. Bundan elektron energiyasi:

$$E_{\max} = m_n c^2 - (m_p + m_{\bar{\nu}})c^2 = (m_n - m_p - m_{\bar{\nu}})c^2$$

Bu sodda ifoda, ammo impuls saqlanishini ham tekshirish kerak. Agar proton va anti-neytrino bir-biriga nisbatan tinch bo'lsa, ular birgalikda massasi $m_p + m_{\bar{\nu}}$ bo'lgan yagona zarracha kabi harakat qiladi. Ular birga harakatlanayotgan bo'lsa, sistemaning

to'liq impulsi nol bo'lishi uchun elektron ularga qarama-qarshi yo'nalishda harakatlanishi kerak va impulsning kattaligi teng bo'lishi shart. Bu holda proton+anti-neytrino tizimining energiyasi $\sqrt{(m_p + m_{\bar{\nu}})^2 c^4 + p^2 c^2}$ bo'lib, bu yerda p — tizimning impulsi. Agar $p = 0$ bo'lsa, u holda ularning energiyasi $(m_p + m_{\bar{\nu}})c^2$ bo'ladi, lekin impuls saqlanishi uchun elektronning impulsi ham nol bo'lishi kerak, bu esa elektronning tinch holatini beradi. Bu holda elektron energiyasi $m_e c^2$ bo'lib, maksimal emas. Demak, maksimal elektron energiyasida proton va anti-neytrino bir-biriga nisbatan tinch emas, balki ularning nisbiy impulsi shunday tanlanadiki, natijada elektron eng katta energiyaga ega bo'ladi. Bu tipik uch jisimli yemirilishda, elektron energiyasining maksimumi anti-neytrino massasi nolga teng bo'lganda $E_{\max} = \frac{(m_n^2 + m_e^2 - m_p^2)c^2}{2m_n}$ formula bilan beriladi (neytrino massasi nol deb hisoblab). Ammo bu yerda neytrino massasi kichik, lekin noldan farqli. Aniqroq ifoda:

Uch jisimli yemirilishda $E_{\max}$ (elektron energiyasining maksimali) qolgan ikki zarracha (proton va anti-neytrino) bir xil tezlik bilan bir yo'nalishda harakatlanganda (ya'ni ular bir-biriga nisbatan tinch holatda) erishiladi. Bu holda ularning umumiy impulsi $\vec{p}_p + \vec{p}_{\bar{\nu}} = \vec{P}$ va umumiy energiyasi $E_p + E_{\bar{\nu}} = \sqrt{(m_p c^2)^2 + (p_p c)^2} + \sqrt{(m_{\bar{\nu}} c^2)^2 + (p_{\bar{\nu}} c)^2}$. Agar ular bir-biriga nisbatan tinch bo'lsa, $\vec{p}_p = \frac{m_p}{m_p + m_{\bar{\nu}}} \vec{P}$ va $\vec{p}_{\bar{\nu}} = \frac{m_{\bar{\nu}}}{m_p + m_{\bar{\nu}}} \vec{P}$ (norelativistik emas, relativistik to'g'ri munosabat: tezliklari bir xil bo'lgani uchun $\vec{p}_p = \frac{E_p}{E_p + E_{\bar{\nu}}} \vec{P}$, lekin tinch massalari nisbati bilan oddiy emas). To'g'ri yondashuv: Ikki zarracha bir xil tezlikka ega bo'lganda, ularning to'rt-impulslari yig'indisi $(E_p + E_{\bar{\nu}}, \vec{P})$ invariant massasi $M_{p\bar{\nu}} = \sqrt{(E_p + E_{\bar{\nu}})^2/c^2 - P^2}$ ga teng va bu ularning tinch massalari yig'indisidan katta (chunki ular birgalikda harakatlanadi). Elektron energiyasi maksimal bo'lganda, $M_{p\bar{\nu}}$ minimal qiymatga erishadi. Minimal invariant massa $m_p + m_{\bar{\nu}}$ ga teng (ikki zarracha bir-biriga nisbatan tinch bo'lganda). Bu holatda $P = 0$ va $E_p + E_{\bar{\nu}} = (m_p + m_{\bar{\nu}})c^2$ bo'ladi. Ammo impuls saqlanishidan $P = p_e$ (elektron impulsi). Demak, $p_e = 0$ bo'lib, elektron tinch holatda bo'ladi — bu minimal elektron energiyasiga olib keladi. Shuning uchun maksimum boshqa shartda: proton va anti-neytrino bir xil yo'nalishda harakatlanadi, lekin ularning nisbiy tezligi nol emas. Ma'lumki, uch jisimli yemirilishda elektron energiyasining maksimumi, qolgan ikki zarracha bir-biriga parallel ravishda bir xil tezlik bilan harakatlanganda (ya'ni ularning nisbiy tezligi nol) erishiladi. Bu holda ular bir-biriga nisbatan tinch bo'lgani uchun ularning invariant massasi $m_p + m_{\bar{\nu}}$ bo'ladi. U holda energiya saqlanishi:

$$E_e + \sqrt{(m_p + m_{\bar{\nu}})^2 c^4 + p_e^2 c^2} = m_n c^2$$

Impuls saqlanishi: $p_e = p_{p\bar{\nu}}$. Bu yerda p_e — elektron impulsi, $p_{p\bar{\nu}}$ — proton va anti-neytrino sistemasining impulsi. Ammo bu sistema tinch markazida ularning umumiy energiyasi $(m_p + m_{\bar{\nu}})c^2$, shuning uchun laboratoriya sistemasida ularning energiyasi $\sqrt{(m_p + m_{\bar{\nu}})^2 c^4 + p_{p\bar{\nu}}^2 c^2}$. Shunday qilib, bizda:

$$E_e + \sqrt{(m_p + m_{\bar{\nu}})^2 c^4 + p_e^2 c^2} = m_n c^2$$

va $E_e^2 = m_e^2 c^4 + p_e^2 c^2$. Bu tenglamalarni yechib E_e ni topamiz. Bu standart natijaga olib keladi:

$$E_{\max} = \frac{(m_n^2 + m_e^2 - (m_p + m_{\bar{\nu}})^2)c^2}{2m_n}$$

Agar $m_{\bar{\nu}}$ juda kichik bo'lsa, uni $m_{\bar{\nu}} = 0$ deb hisoblab, yaqinlashish mumkin. Berilgan qiymatlar bilan hisoblaymiz.

Avval m_n, m_p, m_e ni MeV/c^2 da berilgan. $m_{\bar{\nu}} < 7.3 \text{ eV}/c^2 = 7.3 \times 10^{-6} \text{ MeV}/c^2$. Shuning uchun $m_{\bar{\nu}}$ ni e'tiborsiz qoldirish mumkin. U holda:

$$E_{\max} \approx \frac{(m_n^2 + m_e^2 - m_p^2)c^2}{2m_n}$$

Avval m_n^2, m_p^2, m_e^2 ni hisoblaymiz (MeV^2/c^2 da):

$$m_n^2 = (939.56563)^2 = 882787.5 \quad (\text{aniqroq: } 939.56563^2 = 882787.9)$$

$$m_p^2 = (938.27231)^2 = 880354.5$$

$$m_e^2 = (0.5109907)^2 = 0.261112$$

Shuning uchun $m_n^2 + m_e^2 - m_p^2 = 882787.9 + 0.2611 - 880354.5 = 2433.66 \text{ MeV}^2/c^4$

$$E_{\max} = \frac{2433.66}{2 \times 939.56563} \text{ MeV} = \frac{2433.66}{1879.13126} = 1.295 \text{ MeV}$$

Bu elektronning kinetik energiyasiga o'xshaydi, lekin to'liq energiya. Elektronning tinch massasi 0.511 MeV , shuning uchun kinetik energiya $E_{\max} - m_e c^2 \approx 1.295 - 0.511 = 0.784 \text{ MeV}$. Bu odatdagi beta-yemirilishdagi maksimal elektron energiyasiga mos (0.782 MeV). Biroq bizga to'liq energiya $E_{\max}$ kerak.

Endi $m_{\bar{\nu}}$ ni hisobga olsak, $(m_p + m_{\bar{\nu}})^2 = m_p^2 + 2m_p m_{\bar{\nu}} + m_{\bar{\nu}}^2 \approx m_p^2 + 2m_p m_{\bar{\nu}}$. Shuning uchun

$$E_{\max} = \frac{m_n^2 + m_e^2 - m_p^2 - 2m_p m_{\bar{\nu}} - m_{\bar{\nu}}^2}{2m_n} c^2 \approx E_{\max}^{(0)} - \frac{m_p}{m_n} m_{\bar{\nu}} c^2$$

$m_p/m_n \approx 0.9986$, $m_{\bar{\nu}} < 7.3 \times 10^{-6} \text{ MeV}/c^2$, shuning uchun tuzatish $< 7.3 \times 10^{-6} \text{ MeV}$, bu 1.295 MeV ga nisbatan juda kichik. Demak, $E_{\max} \approx 1.295 \text{ MeV}$ (aniqroq: 1.292 MeV ? Keling, aniq hisoblaymiz).

Berilgan qiymatlar bilan:

$$m_n = 939.56563, \quad m_p = 938.27231, \quad m_e = 0.5109907$$

$$m_n^2 = 939.56563^2 = 882787.548, \quad m_p^2 = 938.27231^2 = 880353.748, \quad m_e^2 = 0.261110$$

$$m_n^2 + m_e^2 - m_p^2 = 882787.548 + 0.26111 - 880353.748 = 2434.06111$$

$$E_{\max}^{(0)} = \frac{2434.06111}{2 \times 939.56563} = \frac{2434.06111}{1879.13126} = 1.29526 \text{ MeV}$$

Demak, $E_{\max} = 1.295 \text{ MeV}$ (3 ta muhim raqam: 1.30 MeV ? 1.295 — 3 ta raqam 1.30 emas, chunki 1.295 da 4 ta raqam. 3 ta muhim raqam: 1.30 MeV . Ammo odatda bu 1.293 MeV deb beriladi. Keling, 1.30 MeV yozaylik.

Anti-neytrino tezligi v_m ga kelsak: $E = E_{\max}$ bo'lganda, anti-neytrino energiyasi minimal bo'ladi? Aslida, maksimal elektron energiyasida anti-neytrino energiyasi minimal bo'ladi. Ammo masalada "speed of the anti-neutrino when $E = E_{\max}$ " so'ralgan. U holda anti-neytrino energiyasi $E_{\bar{\nu}} = m_n c^2 - E_{\max} - E_p$, ammo buni topish uchun proton energiyasini bilish kerak. Yana bir usul: Maksimal elektron energiyasida proton va anti-neytrino bir-biriga parallel va bir xil tezlikda harakatlanadi (ularning nisbiy tezligi nol). U holda ularning umumiy impulsi $p_{p\bar{\nu}} = p_e$ (elektron impulsi). Elektron energiyasi $E_{\max}$ bo'lganda, $p_e c = \sqrt{E_{\max}^2 - m_e^2 c^4}$. Proton va anti-neytrino

tizimining umumiy energiyasi $E_{p\bar{\nu}} = m_n c^2 - E_{\max}$. Ularning invariant massasi $M = \sqrt{E_{p\bar{\nu}}^2/c^2 - p_e^2}$. Bu $M = m_p + m_{\bar{\nu}}$ ga teng bo'lishi kerak (chunki ular bir-biriga nisbatan tinch). Bu shart yordamida $E_{\max}$ ni topdik. Endi anti-neytrinoning o'z tezligini topish uchun, proton va anti-neytrino bir xil tezlikka ega. U holda ularning tezligi v ga teng. Ularning umumiy energiyasi $E_{p\bar{\nu}} = \gamma(m_p + m_{\bar{\nu}})c^2$, umumiy impulsi $p_{p\bar{\nu}} = \gamma(m_p + m_{\bar{\nu}})v$. Shuningdek, $p_{p\bar{\nu}} = p_e$. Demak, $v = \frac{p_e c^2}{E_{p\bar{\nu}}}$. Anti-neytrino tezligi ham shu v ga teng. Shuning uchun:

$$v_m = c \cdot \frac{p_e c}{E_{p\bar{\nu}}} = c \cdot \frac{\sqrt{E_{\max}^2 - m_e^2 c^4}}{m_n c^2 - E_{\max}}$$

Hisoblaymiz: $E_{\max} = 1.29526 \text{ MeV}$, $m_e c^2 = 0.51099 \text{ MeV}$, shuning uchun

$$E_{\max}^2 = 1.6776, \quad m_e^2 = 0.26111, \quad E_{\max}^2 - m_e^2 = 1.4165, \quad \sqrt{} = 1.1902 \text{ MeV}$$

$$E_{p\bar{\nu}} = m_n c^2 - E_{\max} = 939.56563 - 1.29526 = 938.27037 \text{ MeV}$$

$$\frac{v_m}{c} = \frac{1.1902}{938.27037} = 0.0012685 \approx 1.27 \times 10^{-3}$$

3 ta muhim raqam: $v_m/c = 1.27 \times 10^{-3}$.

Agar $m_{\bar{\nu}}$ ni hisobga olsak, u juda kichik bo'lgani uchun bu natija o'zgarishsiz qoladi.

Shunday qilib, javoblar:

$$E_{\max} = \frac{(m_n^2 + m_e^2 - (m_p + m_{\bar{\nu}})^2)c^2}{2m_n} \approx \frac{(m_n^2 + m_e^2 - m_p^2)c^2}{2m_n} = 1.30 \text{ MeV}$$

$$\frac{v_m}{c} = \frac{\sqrt{E_{\max}^2 - m_e^2 c^4}}{m_n c^2 - E_{\max}} \approx 1.27 \times 10^{-3}$$

Javoblar varaqasi (3-masala)

$$E_{\max} = \frac{(m_n^2 + m_e^2 - (m_p + m_{\bar{\nu}})^2)c^2}{2m_n} = 1.30 \text{ MeV}$$

$$\frac{v_m}{c} = \frac{\sqrt{E_{\max}^2 - m_e^2 c^4}}{m_n c^2 - E_{\max}} = 1.27 \times 10^{-3}$$

Nazariy masala 3 (davomi)

B qism: Yorug'lik bilan levitatsiya (Light Levitation)

Shaffof shisha yarimsharning radiusi R , massasi m va sindirish ko'rsatkichi (index of refraction) n . Yarimshardan tashqaridagi muhitda sindirish ko'rsatkichi birga teng. Monoxromatik lazer nurlarining parallel dastasi 3-rasmda ko'rsatilganidek, uning tekis sirtining markaziy qismiga tekis va normal tushadi. Og'irlik tezlanishi $\vec{g}$ vertikal pastga yo'nalgan. Lazer dastasining aylana ko'ndalang kesimi radiusi δ va u R dan ancha kichik. Shisha yarimshar ham, lazer dastasi ham z -o'qiga nisbatan o'q simmetrik (axially symmetric).

Shisha yarimshar hech qanday lazer nurini yutmaydi. Uning sirti yupqa shaffof material bilan qoplangan, shuning uchun yorug'lik yarimsharga kirish va chiqishda qaytish (reflections) ahamiyatsiz. Yorug'likning qaytarmaydigan sirt qatlamidan o'tadigan optik yo'li ham ahamiyatsiz.

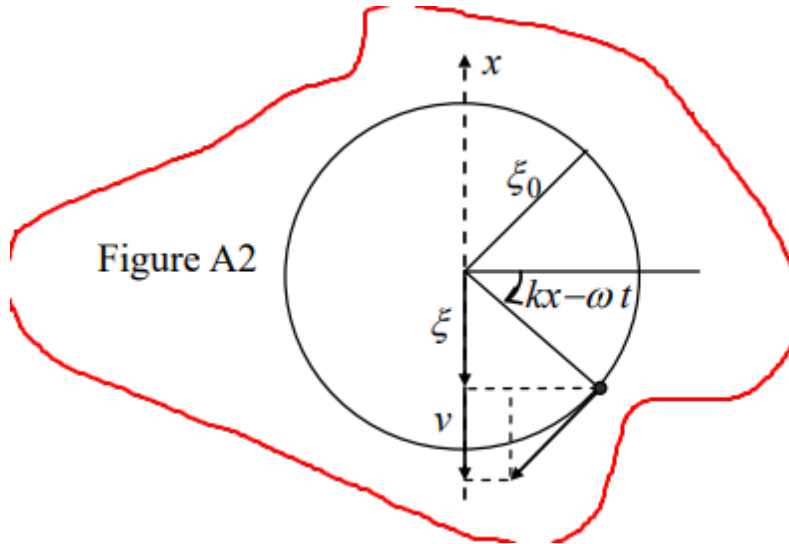


Figure 17: 3-rasm (Figure 3)

- (b) $(\delta/R)^3$ yoki undan yuqori tartibli hadlarni e'tiborsiz qoldirib, shisha yarimsharning og'irligini muvozanatlashtirish uchun zarur bo'lgan lazer quvvati P ni toping. (4.0 ball)

Ko'rsatma: θ birdan ancha kichik bo'lganda, $\cos \theta \approx 1 - \theta^2/2$.

3-masala yechimi (davomi)

B qism: Yorug'lik bilan levitatsiya

(b) B1-rasmga murojaat qiling. Sferik sirtga yorug'likning sinishi (refraction) Snellius qonuniga bo'ysunadi:

$$n \sin \theta_i = \sin \theta_t \quad (1B)$$

$(\delta/R)^3$ yoki undan yuqori tartibli hadlarni sinus funksiyalarda e'tiborsiz qoldirib, (1B) tenglama quyidagicha bo'ladi:

$$n\theta_i \approx \theta_t \quad (2B)$$

B1-rasmdagi $\triangle FAC$ uchburchak uchun:

$$\beta = \theta_t - \theta_i \approx n\theta_i - \theta_i = (n - 1)\theta_i \quad (3B)$$

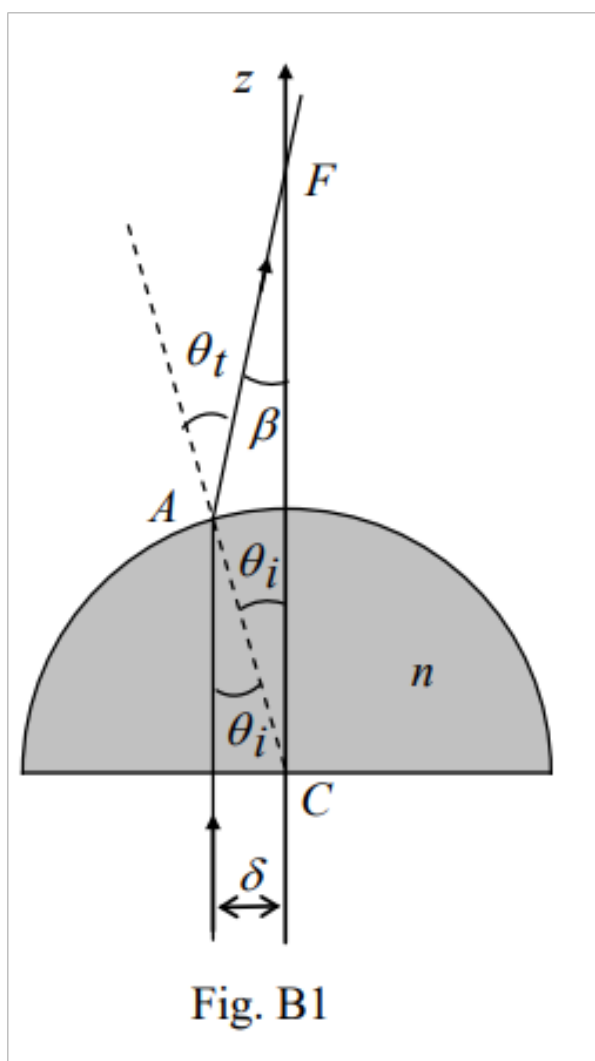


Figure 18: B1-rasm (Figure B1)

Tushayotgan yorug'lik chastotasi f_0 bo'lsin. Agar tekis sirtga birlik yuzaga birlik vaqtda tushayotgan fotonlar soni n_p bo'lsa, u holda tekis sirtga birlik vaqtda tushayotgan fotonlarning to'liq soni $n_p \pi \delta^2$ ga teng. Tekis sirtga tushayotgan fotonlarning

to'liq quvvati $(n_p \pi \delta^2)(hf_0)$, bu yerda h — Plank doimiysi. Demak,

$$n_p = \frac{P}{\pi \delta^2 h f_0} \quad (4B)$$

Tekis sirtida ichki radiusi r va tashqi radiusi $r + dr$ bo'lgan halqasimon diskka birlik vaqtda tushayotgan fotonlar soni $n_p(2\pi r dr)$ ga teng, bu yerda $r = R \tan \theta_i \approx R \theta_i$. Shuning uchun,

$$n_p(2\pi r dr) \approx n_p(2\pi R^2) \theta_i d\theta_i \quad (5B)$$

Bu fotonlar sferik sirtida sinishidan keyin olib ketilgan vaqt birligidagi impulsning (momentum) z -tashkil etuvchisi:

$$\begin{aligned} dF_z &= n_p \frac{hf_0}{c} (2\pi r dr) \cos \beta \approx n_p \frac{hf_0}{c} (2\pi R^2) \left(1 - \frac{\beta^2}{2}\right) \theta_i d\theta_i \\ &\approx n_p \frac{hf_0}{c} (2\pi R^2) \left[\theta_i - \frac{(n-1)^2}{2} \theta_i^3\right] d\theta_i \end{aligned} \quad (6B)$$

Shuning uchun vaqt birligida olib ketilgan to'liq impulsning z -tashkil etuvchisi:

$$\begin{aligned} F_z &= 2\pi R^2 n_p \left(\frac{hf_0}{c}\right) \int_0^{\theta_{im}} \left[\theta_i - \frac{(n-1)^2}{2} \theta_i^3\right] d\theta_i \\ &= \pi R^2 n_p \left(\frac{hf_0}{c}\right) \theta_{im}^2 \left[1 - \frac{(n-1)^2}{4} \theta_{im}^2\right] \end{aligned} \quad (7B)$$

bu yerda $\tan \theta_{im} = \frac{\delta}{R} \approx \theta_{im}$. Shunday qilib, (4B) natijasiga ko'ra:

$$F_z = \frac{\pi R^2 P}{\pi \delta^2 h f_0} \left(\frac{hf_0}{c}\right) \frac{\delta^2}{R^2} \left[1 - \frac{(n-1)^2 \delta^2}{4R^2}\right] = \frac{P}{c} \left[1 - \frac{(n-1)^2 \delta^2}{4R^2}\right] \quad (8B)$$

Yorug'lik levitatsiyasi kuchi (optical levitation) shisha yarimsharga tushayotgan va sinayotgan yorug'lik tomonidan qo'yilgan kuchlarning z -tashkil etuvchilari yig'indisiga teng va quyidagicha beriladi:

$$\frac{P}{c} + (-F_z) = \frac{P}{c} - \frac{P}{c} \left[1 - \frac{(n-1)^2 \delta^2}{4R^2}\right] = \frac{(n-1)^2 \delta^2}{4R^2} \frac{P}{c} \quad (9B)$$

Buni shisha yarimsharning og'irligi mg ga tenglashtirib, yarimsharni levitatsiya qilish uchun zarur minimal lazer quvvati:

$$P = \frac{4mgcR^2}{(n-1)^2 \delta^2} \quad (10B)$$

Baholash sxemasi (Marking Scheme)

Nazariy masala 3

Neytrino massasi va neytron yemirilishi

Umumiy ball	Qism ball	Javoblar uchun baholash mezonlari
8*4.0 ball	8*(a) 4.0	Elektronning maksimal energiyasi va unga mos keluvchi anti-neytrino tezligi.
		0.5: energiya-impuls saqlanishidan foydalanadi va ularni tenglamalarga aylantira oladi.
		0.5: E_e uchun uning maksimal qiymatini aniqlash imkonini beradigan ifodani oladi.
		(0.5+0.2): E_e maksimal bo'lganda proton va anti-neytrino bir xil tezlik bilan harakatlanishi kerak degan xulosaga keladi (0.2 bir xil yo'nalish uchun).
		0.6: $(E_p E_v - \vec{q}_p \cdot \vec{q}_v)$ ning minimal qiymati $m_p m_v$ ga teng ekanligini yoki unga ekvivalent xulosani o'rnatadi.
		(0.5+0.1): $E_{\max}$ ifodasi va qiymati.
		0.5: $\beta_v = \sqrt{E_e^2 - m_e^2}/(m_n - E_e)$ ekanligini xulosa qiladi.
		(0.5+0.1): v_m/c ifodasi va qiymati.

Yorug'lik levitatsiyasi (Light Levitation)

Umumiy ball	Qism ball	Javoblar uchun baholash mezonlari
12*4.0 ball	12*(b) 4.0	Shisha yarimsharning og'irligini muvozanatlash uchun zarur lazer quvvati.
		0.3: sinish qonuni $n \sin \theta_i = \sin \theta_t$.
		0.3: chiziqli yaqinlashish $n \theta_i \approx \theta_t$ ni qo'llaydi.
		0.4: og'ish va tushish burchaklari orasidagi munosabat.
		0.3: foton energiyasi $\epsilon = h\nu$.
		0.3: foton impulsi $p = \epsilon/c$.
		0.3: vaqt birligida tushayotgan fotonlarning impulsi $= P/c$.
		0.6: tushish burchagi funksiyasi sifatida vaqt birligida sinib chiqqan fotonlarning impulsi.
		0.4: vaqt birligida sinib chiqqan fotonlarning to'liq impulsi $= [1 - (n - 1)^2 \delta^2 / (4R^2)] P/c$.
		0.4: levitatsiya kuchi = tushayotgan va sinayotgan fotonlar ta'sir kuchlarining yig'indisi.
		0.4: levitatsiya kuchi $= (n - 1)^2 \delta^2 P / (4cR^2)$.
		0.3: zarur lazer quvvati $P = 4mgcR^2 / (n - 1)^2 \delta^2$.

Eksperimental tanlov (Experimental Competition)

34-Xalqaro fizika olimpiadasi, Taypey, Tayvan

Chorshanba, 2003 yil 6 avgust Vaqt: 5 soat

Avval buni o'qing:

- (a) Faqat berilgan qalamdan foydalaning.
- (b) Javob varaqlari va qog'ozlarning faqat old tomonidan foydalaning.
- (c) Javoblaringizda iloji boricha kam matn ishlatib; asosan tenglamalar, raqamlar va rasmlar bilan ifodalang. Agar kerakli natija sonli qiymat bo'lsa, yakuniy natijangizni to'liqsimon chiziq bilan tagiga chizing.
- (d) O'lchov natijalari va masala yechimi uchun zarur deb hisoblagan, baholanishini xohlagan barcha narsalarni bo'sh qog'oz varaqlariga yozing.
- (e) Har bir ishlatilgan qog'oz varag'ining yuqori qismidagi katakchalarga o'z mamlakatingiz va o'quvchi raqamingizni [Student No.] yozish mutlaqo zarur. Bundan tashqari, har bir masala uchun ishlatilgan bo'sh qog'oz varaqlariga masala raqamini [Question No.: masalan A-(1)], har bir varaqning tartib raqamini [Page No.] va har bir masala uchun ishlatgan va baholanishini xohlagan bo'sh varaqlarning umumiy sonini [Total No. of pages] yozing. Agar siz baholanishini xohlamagan eslatmalar uchun bo'sh qog'oz varaqlaridan foydalansangiz, butun varaq bo'ylab katta X chizing va ularni raqamlashingizga qo'shmang.
- (f) Imtihon oxirida javob varaqlaringiz va grafiklaringizni tartibli joylashtiring.
- (g) Grafiklardagi xato chiziqlari (error bars) faqat eksperimentning A qismida kerak.
- (h) **Ehtiyot bo'ling: Lazer nuriga to'g'ridan-to'g'ri qaramang. Ko'zingizga zarar yetishi mumkin!!**

Asboblar va materiallar

Table 3: Mavjud asboblar va materiallar

T/r	Asbob va material	
A	Fotodetektor (PD)	
B	Aylanadigan moslamali polarizatorlar	
C	90° TN-LC hujayra (sariq simlar) aylanadigan LC moslamasi bilan	
D	Funksiya generatori	
E	Lazer diodi (LD)	
F	Multimetrlar	
G	Parallel LC hujayra (apelsin simlar)	
H	O'zgaruvchan qarshilik (rezistor)	
I	Batareyalar	
J	Batareya qutisi	
K	Optik skameyka	
L	Qisman shaffof qog'ozlar	
M	Chizg'ich	
N	Oq lenta* (asboblarni belgilash uchun)	
O	Qaychi	
P	Grafik qog'ozlari	

* Asboblarga to'g'ridan-to'g'ri belgi qo'ymang. Kerak bo'lganda, qismlarga bir parcha oq lentani yop

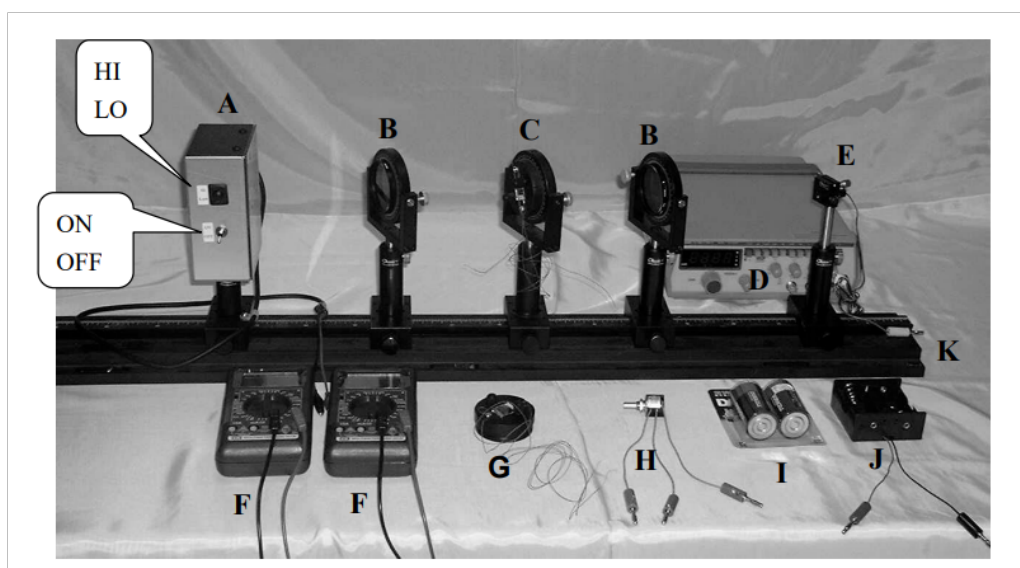


Figure 19: 1-rasm (Fig. 1)

Javoblar varaqasi (3-masala, B qism)

$$(b) P = \frac{4mgcR^2}{(n-1)^2\delta^2}$$

Eksperimental masala (davomi)

2. Multimetr uchun ko'rsatmalar:

- “DC/AC” tugmasi — o'zgarmas yoki o'zgaruvchan tokni o'lchashni tanlash uchun.
- “V” va “COM” kirishlari — kuchlanish va qarshilik o'lchash uchun.
- “mA” va “COM” kirishlari — kichik tokni o'lchash uchun. Display tokni milliamperlarda ko'rsatadi.
- Funksiya tugmasi — kerakli funksiya va o'lchov oralig'ini tanlash uchun. “V” — kuchlanish, “A” — tok, “Ω” — qarshilik.

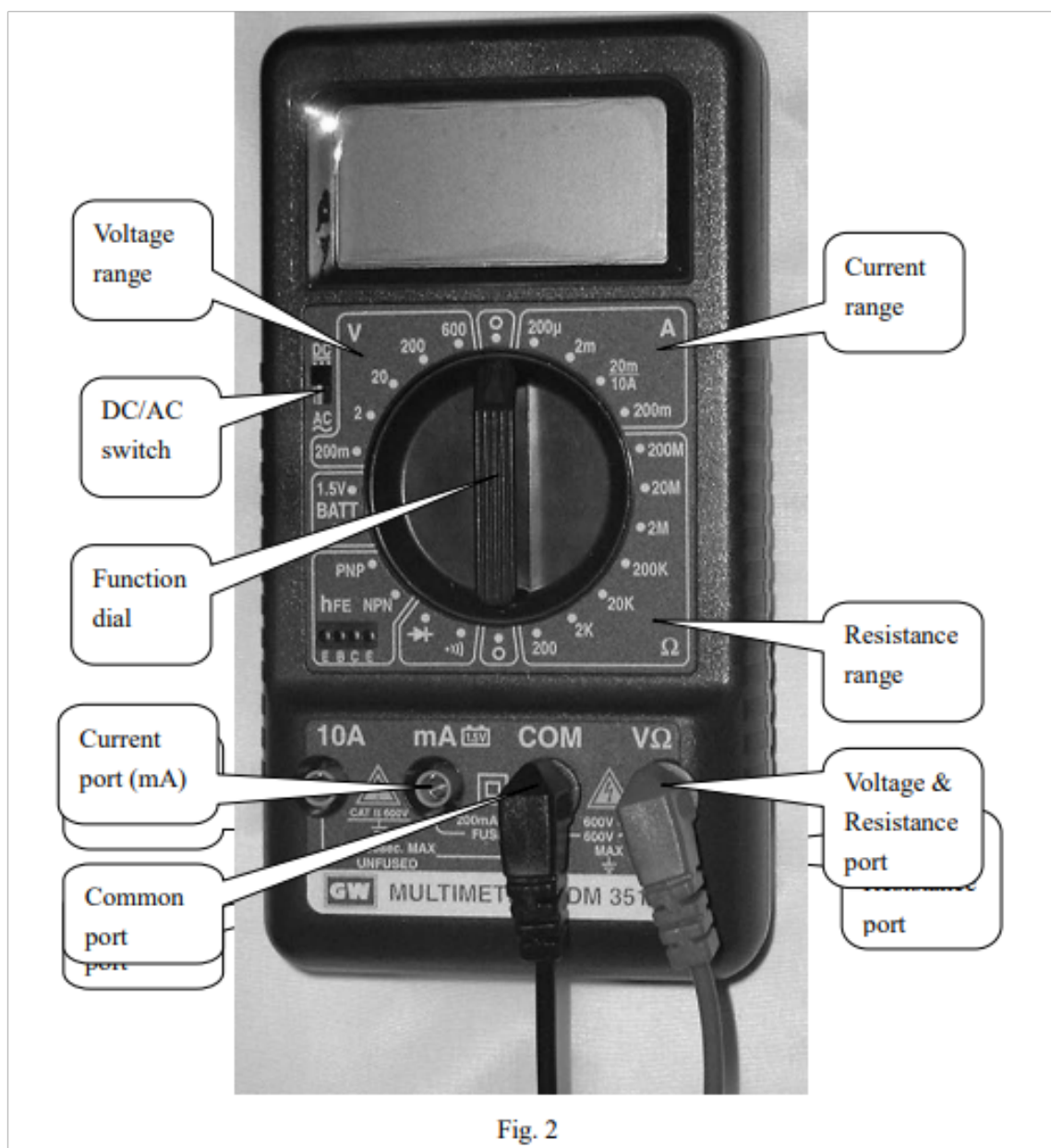


Fig. 2

Figure 20: 2-rasm (Fig. 2)

3. Funksiya generatori uchun ko'rsatmalar:

- Quvvat tugmasi — bir marta bosilsa yoqiladi, yana bosilsa o'chadi.
- Chastota diapazonini tanlang va tegishli tugmani bosing.
- Chastota raqamli displeyda ko'rsatiladi.
- Chastotani aniq sozlash uchun qo'pol va nozik sozlash tugmalaridan foydalaning.
- Eng chap to'lqin shakli tugmasini bosib, to'g'ri to'rtburchak (square-wave) shaklini tanlang.
- Chiqish kuchlanishini o'zgartirish uchun amplituda sozlash tugmasidan foydalaning.

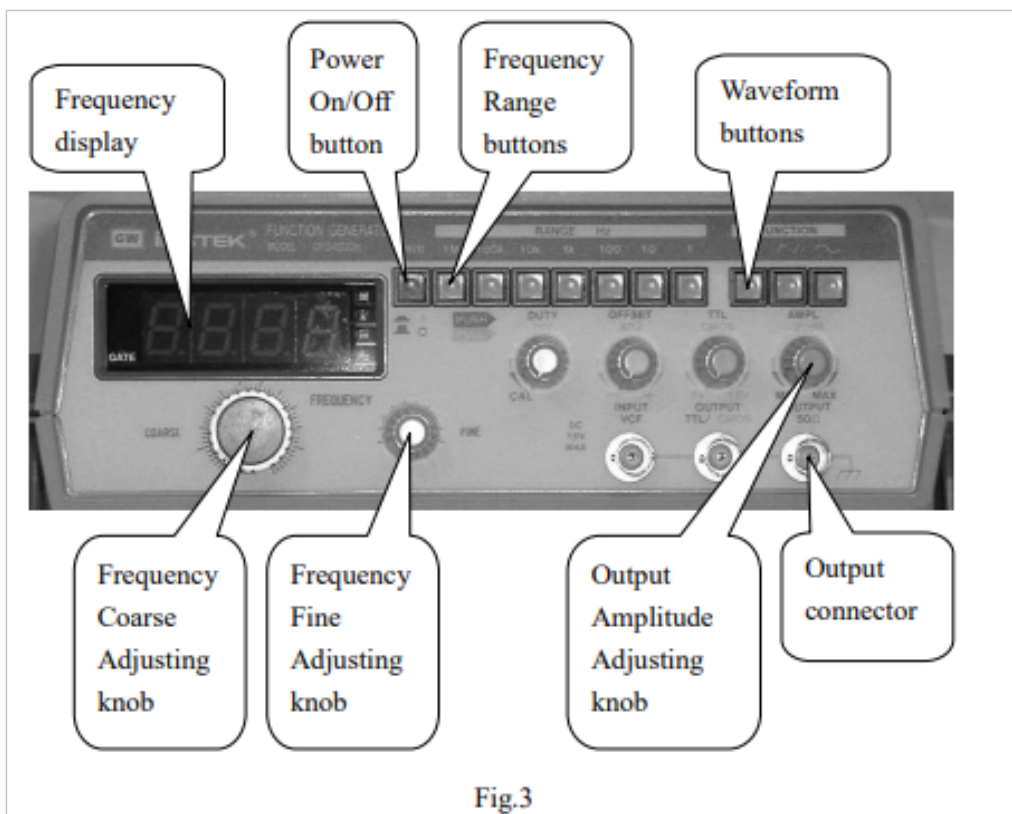


Fig.3

Figure 21: 3-rasm (Fig. 3)

A qism: Lazer diodining optik xususiyatlari

I. Kirish

1. Lazer diodi

Ushbu tajribada yorug'lik manbai — to'lqin uzunligi 650 nm bo'lgan lazer nurini chiqaradigan lazer diodi (LD). Lazer diodining toki chegara tokidan (threshold current) katta bo'lganda, u monoxromatik, qisman qutblangan va kogerent nurlanish chiqaradi. Agar lazer diodining toki chegara tokidan kichik bo'lsa, chiqarilgan nur intensivligi juda kichik. Chegara tokidan yuqorida nur intensivligi tok bilan keskin ortadi va tok bilan chiziqli munosabatda bo'ladi. Agar tok yanada oshirilsa, lazer diodining temperaturasi ko'tarilishi tufayli intensivlikning tok bo'yicha o'sish tezligi kamayadi. Shuning uchun lazer diodi uchun optimal ish oralig'i — intensivlik tok bilan chiziqli bo'lgan soha. Odatda chegara toki I_{th} — chiziqli sohaning ekstrapolyatsiya chizig'ining tok o'qi bilan kesishish nuqtasi sifatida aniqlanadi.

Ehtiyot bo'ling: Lazer nuriga to'g'ridan-to'g'ri qaramang. Ko'zingizga zarar yetishi mumkin!!

2. Fotodetektor

Ushbu tajribada ishlatiladigan fotodetektor fotodiod va tok kuchaytirgichdan iborat. Fotodiodga tashqi kuchlanish berilganda, diodga tushgan yorug'lik ta'sirida fototok hosil bo'ladi. O'zgarmas temperatura va monoxromatik tushuvchi yorug'lik

sharoitida fototok yorug'lik intensivligiga proporsionaldir. Tok kuchaytirgich fototokni chiqish kuchlanishiga aylantiradi. Bizning fotodetektorimizda ikkita uzatish koeffitsienti mavjud — yuqori va past kuchaytirish. Tajribada faqat past kuchaytirish ishlatiladi. Biroq, fotodiodning o'z cheklovi tufayli, agar yorug'lik intensivligi juda yuqori bo'lsa, chiqish kuchlanishi taxminan 8 Vol'tda to'yinishga (saturation) erishadi va fotodiod to'g'ri ishlay olmaydi. Shuning uchun fotodetektorning to'g'ri ish oralig'i — chiqish kuchlanishi yorug'lik intensivligiga proporsional bo'lgan sohadir. Agar yorug'lik intensivligi juda yuqori bo'lib, fotodiod to'yinishga erishsa, fotodetektor ko'rsatkichi tushayotgan yorug'lik intensivligini to'g'ri aks ettira olmaydi.

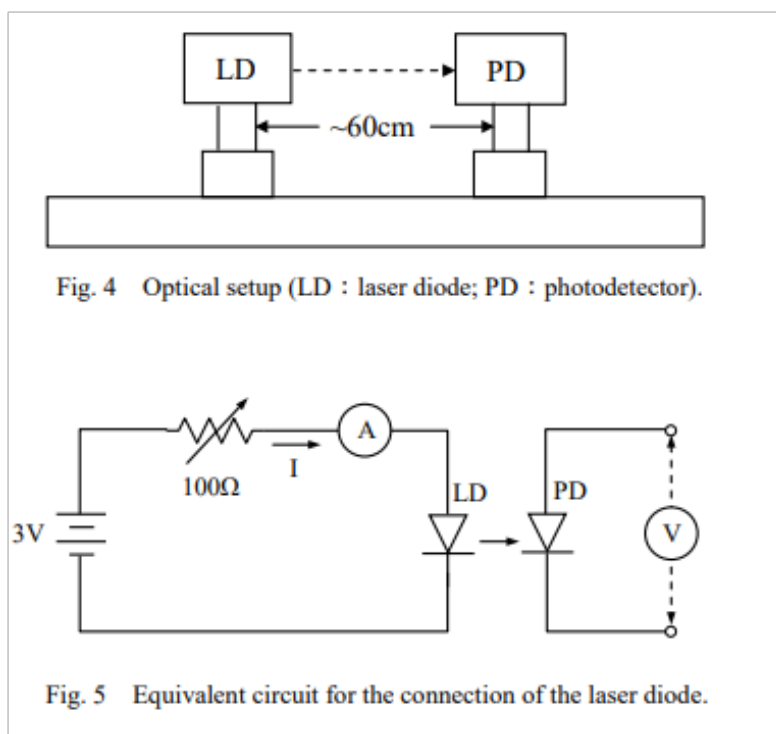
II. Tajribalar va jarayonlar

Lazer diodi va fotodetektorning xarakteristikalar

Tajribalar muvaffaqiyatli o'tkazilishi uchun tajriba qurilmasining turli qismlari orasidagi optik moslik (optical alignment) juda muhim. Shuningdek, yorug'lik manbai va detektor to'g'ri sharoitda ishlatilishi kerak. A qism ana shu masalalarga va qutblanish darajasi masalasiga bag'ishlangan.

- (a) Lazer diodi va fotodetektorni optik skameykada gorizontall chiziq bo'ylab o'rnatish (4-rasm). O'zgaruvchan qarshilik, batareyalar to'plami, ampermetr, voltmeter, lazer diodi va fotodetektorni 5-rasmga muvofiq ulang. O'zgaruvchan qarshilikni shunday sozlangki, LD dan o'tayotgan tok 25 mA atrofida bo'lsin va lazer diodi to'g'ri nurlansin. Fotodetektor uchun past kuchaytirishni tanlang. Lazer diodi va fotodetektorni shunday moslangki, lazer nuri detektor qutisidagi kichik teshikka aniq tushsin va fotodetektor ko'rsatkichi maksimal qiymatga erishsin.

Ehtiyot: Qisqa tutashuvni oldini olish uchun batareyalarning qora va qizil simlarini bir-biriga tekkizib qo'ymang.



[H] 4-rasm. Optik qurilma (LD: lazer diodi; PD: fotodetektor), 5-rasm. Lazer diodini ulash uchun

ekvivalent sxema..

- (b) Fotodetektorning chiqish kuchlanishidan lazer nuri intensivligi J ni ifodalash uchun foydalaning. O'zgaruvchan qarshilikni sozlab, lazer diodining toki I ni noldan maksimal qiymatgacha o'zgartiring va I ortishi bilan J ni o'lchang. O'lchashda tokning mos o'sish qadamlarini tanlang.

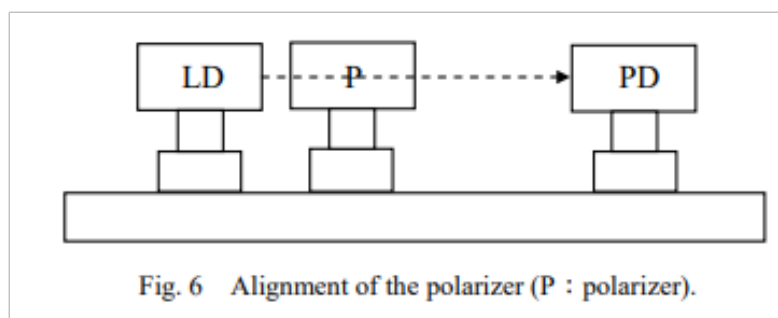
A-(1) savol (1.5 ball)

J ning I ga bog'liqlik jadvalini tuzing, grafigini chizing va unga egri chiziqni torting.

A-(2) savol (3.5 ball)

J vs I egri chizig'ining chiziqli sohasidagi maksimal tok I_m ni va uning xatoligini baholang. Chiziqli sohani $J - I$ grafigida strelkalar ($\downarrow$) bilan belgilang va chegara toki I_{th} ni xatoligi bilan aniqlang.

- (c) Lazer diodining tokini $I_{th} + \frac{2}{3}(I_m - I_{th})$ qilib tanlang, bu lazer diodi va fotodetektorning yaxshi ishlashini ta'minlaydi.
- (d) B qism tajribasiga tayyorgarlik: 6-rasmda ko'rsatilganidek, lazer diodiga yaqin joyga optik skameykaga birinchi polarizatorni o'rnating. Lazer nuri polarizatorning markaziy qismidan o'tishiga ishonch hosil qiling. Polarizatorni shunday sozlangki, tushuvchi lazer nuri polarizator tekisligiga perpendikulyar bo'lsin. (Ko'rsatma: tushuvchi va qaytgan yorug'lik dog'lari bir-biriga mos kelishini tekshirish uchun qisman shaffof qog'ozni sinov ekrani sifatida qo'yishingiz mumkin.)



[H]

6-rasm.

Polarizatorni moslash (P: polarizator).

- (e) Lazer diodining tokini o'zgartirmasdan, ikkinchi polarizatorni optik skameykaga o'rnating va to'g'ri moslashni amalga oshiring — manba, detektor va polarizatorlarni bir to'g'ri chiziqqa qo'ying va har bir polarizator tekisligi yorug'lik nuriga perpendikulyar bo'lishiga ishonch hosil qiling.

B qism: Nematik suyuq kristallarning optik xususiyatlari: 90° TN LC hujayraning elektro-optik kommutatsiya xarakteristikasi

I. Kirish

1. Suyuq kristall

Suyuq kristall (LC) — qattiq kristall va amorf suyuqlik orasidagi oraliq holat. Nematik suyuq kristallar — uzunchoq ignasimon molekulalardan tashkil topgan organik birikmalar. Elektr maydoni qo'llash orqali molekulalarning yo'nalishini osonlik bilan moslash va boshqarish mumkin. Ko'pgina LC asboblarda suyuq kristall molekulalarining bir xil yoki aniq belgilangan yo'nalishi talab qilinadi. Ushbu tajribada ishlatiladigan LC hujayraning tuzilishi 7-rasmda ko'rsatilgan. Poliimid plyonkani ishqalash (rubbing) substrat sirtlarida LC molekulalari uchun yaxshi moslangan afzal yo'nalish hosil qiladi, shuning uchun molekulyar ta'sir tufayli butun LC qatlami bir xil molekulyar yo'nalishga erishadi. Mahalliy molekulyar yo'nalish shu nuqtadagi LC direktori deb ataladi.

LC hujayra ikki asosiy sindirish ko'rsatkichiga ega bo'lgan qo'sh nur sinishi (double refraction) hodisasini namoyish etadi. Yorug'lik direktor yo'nalishi bo'ylab tarqalganda, barcha qutblanish komponentlari bir xil $v_o = c/n_o$ tezlik bilan tarqaladi, bu yerda n_o oddiy sindirish ko'rsatkichi (ordinary index) deb ataladi. Bu tarqalish yo'nalishi (direktor yo'nalishi) LC hujayraning optik o'qi (optic axis) deb ataladi. Yorug'lik dastasi optik o'qqa perpendikulyar yo'nalishda tarqalganda, odatda ikki xil tarqalish tezligi mavjud. Optik o'qqa perpendikulyar (yoki parallel) qutblangan yorug'likning elektr maydoni $v_o = c/n_o$ (yoki $v_e = c/n_e$, bu yerda n_e g'ayrioddiy sindirish ko'rsatkichi (extraordinary index) deb ataladi) tezlik bilan tarqaladi. Ikkilamchi nur sinishi (birefringence) — g'ayrioddiy va oddiy sindirish ko'rsatkichlari ayirmasi sifatida aniqlanadi: $\Delta n \equiv n_e - n_o$.

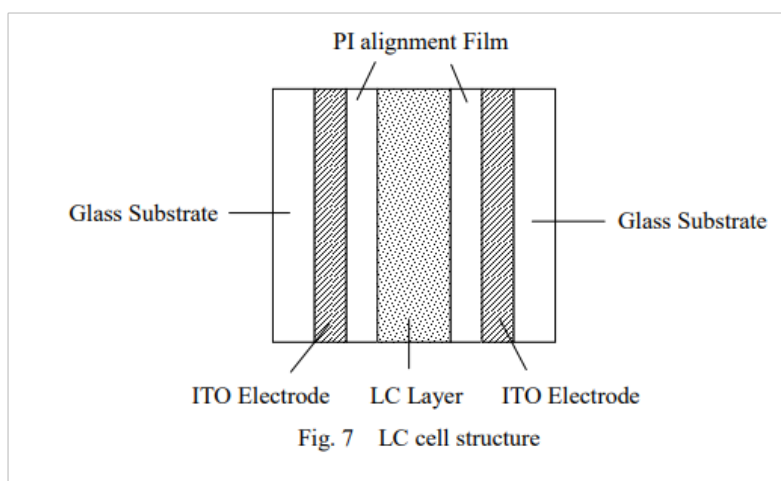


Figure 22: 7-rasm. LC hujayra tuzilishi (Fig. 7).

2. 90° buralgan nematik (TN) hujayra

8-rasmda ko'rsatilgan 90° buralgan nematik (TN) hujayrada, orqa sirtidagi LC direktori oldingi sirtga nisbatan 90° ga buralgan. Oldingi lokal direktor polarizatorning o'tkazish o'qiga parallel qilib o'rnatilgan. Qutblanmagan tushuvchi yorug'lik oldingi polarizator tomonidan chiziqli qutblangan yorug'likka aylantiriladi.

Chiziqli qutblangan yorug'lik 90° TN hujayradan o'tganda, uning qutblanishi LC direktorlarining buralishiga ergashadi (qutblangan yorug'lik faqat n_e ni "ko'radi"), shuning uchun chiqish dastasi chiziqli qutblangan bo'lib qoladi, faqat uning qutblanish o'qi 90° ga buriladi (bu n_e orqali qutblanishni aylantirish effekti deb ataladi; shunga o'xshash n_o orqali ham effekt mavjud). Shunday qilib, 90° TN hujayradan foydalanadigan normal qora (NB) rejimda analizatorning (ikkinchi polarizator) o'tkazish o'qi polarizatorning o'tkazish o'qiga parallel qilib o'rnatiladi (9-rasm). Biroq, LC hujayraga qo'yilgan kuchlanish V kritik qiymat V_c dan oshganda, LC molekulalarining direktori tashqi elektr maydoni (yorug'lik tarqalish yo'nalishi bo'ylab) bo'ylab yo'nalishga intiladi. Shuning uchun LC hujayraning qutblanishni boshqarish effekti asta-sekin yo'qoladi va yorug'lik analizatordan o'ta boshlaydi. Uning elektro-optik kommutatsiya nishabligi $\gamma = (V_{90} - V_{10})/V_{10}$ sifatida aniqlanadi, bu yerda V_{10} va V_{90} — chiqish yorug'lik intensivligi mos ravishda maksimal intensivlikning 10% va 90% ga yetadigan qo'yilgan kuchlanishlardir.

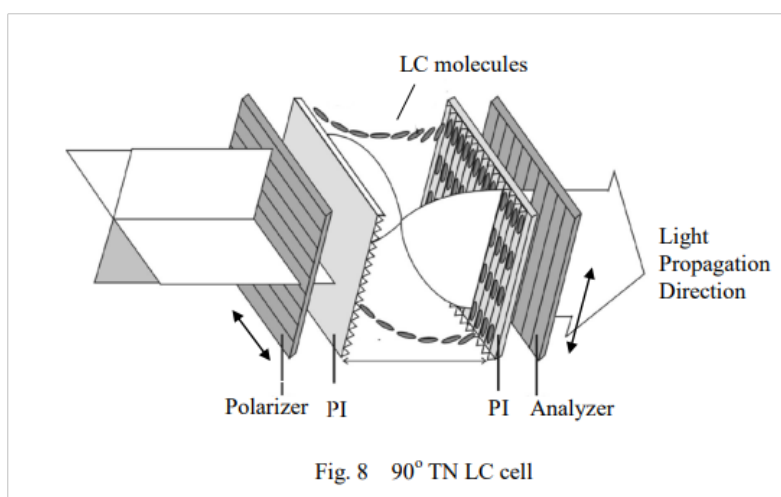


Figure 23: 8-rasm. 90° TN hujayra tuzilishi.

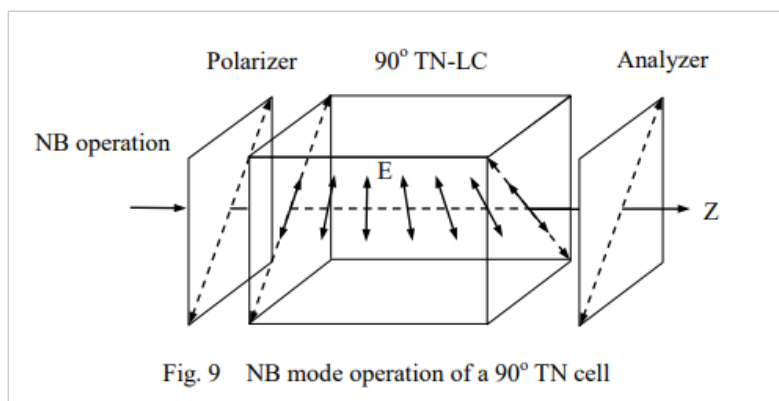


Figure 24: 9-rasm. NB rejimdagi 90° TN hujayraning ishlash prinsipi.

II. Tajribalar va jarayonlar

- (a) Ikki polarizator orasida o'tkazish o'qlari parallel bo'lgan NB 90° TN LC rejimini o'rnatish va funktsiya generatoridan 100 Hz to'g'ri to'rtburchak (square wave) kuchlanishni ikki shisha substratning ITO qismlariga qo'llang va qo'yilgan kuchlanishni (V_{rms}) 0 dan 7.2 Vol'tgacha o'zgartiring.

- Muhim burilish nuqtalarida kerak bo'lsa, ko'proq ma'lumot oling.

B-(1) savol (5.0 ball)

NB 90° TN LC ning elektro-optik kommutatsiya egri chizig'ini (J vs V_{rms}) o'lchang, jadval tuzing, grafigini chizing va uning kommutatsiya nishabligi γ ni toping.

B-(2) savol (2.5 ball)

Ushbu NB 90° TN LC hujayraning kritik kuchlanishi V_c ni aniqlang. V_c qiymatini qanday aniqlaganingizni grafikda aniq ko'rsating. *Ko'rsatma:* Qo'yilgan tashqi kuchlanish kritik kuchlanishdan oshganda, yorug'lik o'tkazuvchanligi tez va keskin ortadi.

C qism: Nematik suyuq kristallarning optik xususiyatlari: parallel yoʻnaltirilgan LC hujayraning elektro-optik kommutatsiya xarakteristikasi

I. Kirish

Parallel yoʻnaltirilgan (homogeneous) LC hujayra

Parallel yoʻnaltirilgan LC hujayrada oldingi va orqa substratlardagi direktorlar bir-biriga parallel (10-rasm). Chiziqli qutblangan yorugʻlik parallel yoʻnaltirilgan hujayraga uning qutblanishi LC direktori (ishqalash yoʻnalishi) boʻylab boʻlganda tushsa, sof faza modulyatsiyasi amalga oshiriladi, chunki yorugʻlik faqat gʻayrioddiy nur (extraordinary ray) sifatida harakat qiladi.

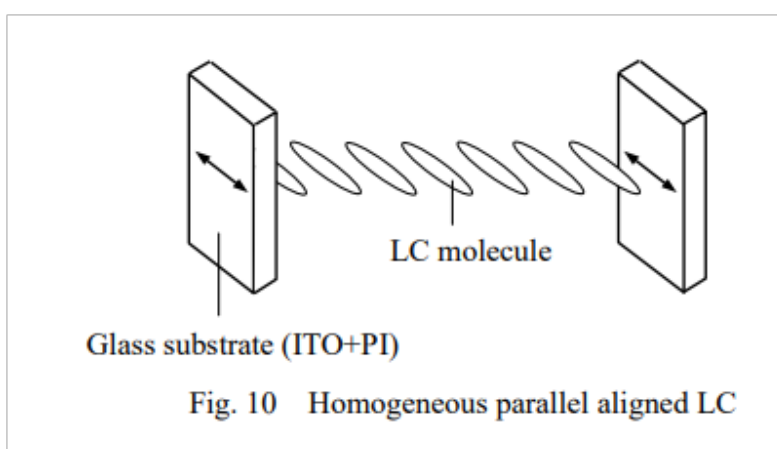


Figure 25: 10-rasm. Bir jinsli parallel yoʻnaltirilgan LC.

Boshqa tomondan, agar chiziqli qutblangan yorugʻlik parallel yoʻnaltirilgan hujayraga uning qutblanishi yoʻnaltirilgan LC direktorlariga nisbatan $\theta = 45^\circ$ burchak ostida (11-rasm) normal tushsa, u holda LC muhitda gʻayrioddiy va oddiy nurlarning har xil tarqalish tezligi tufayli faza kechikishi (phase retardation) yuz beradi. Ushbu $\theta = 45^\circ$ konfiguratsiyada, ikki polarizator parallel boʻlganda, parallel yoʻnaltirilgan LC hujayraning normallashtirilgan oʻtkazuvchanligi quyidagicha ifodalanadi:

$$T_{\parallel} = \cos^2 \frac{\delta}{2}$$

Faza kechikishi δ quyidagicha ifodalanadi:

$$\delta = \frac{2\pi d \Delta n(V, \lambda)}{\lambda}$$

bu yerda d — LC qatlami qalinligi, λ — havodagi yorugʻlik toʻlqin uzunligi, V — qoʻyilgan oʻzgaruvchan kuchlanishning effektiv qiymati (rms), Δn — λ va V ga bogʻliq boʻlgan LC ning ikkilamchi nur sinishi (birefringence). Shuni ham taʼkidlash kerakki, $V = 0$ da $\Delta n (= n_e - n_o)$ oʻzining maksimal qiymatiga ega, shuningdek δ ham maksimal. Δn esa V ortishi bilan kamayadi.

Umumiy holda:

$$T_{\parallel} = 1 - \sin^2 2\theta \sin^2 \frac{\delta}{2}$$

$$T_{\perp} = \sin^2 2\theta \sin^2 \frac{\delta}{2}$$

bu yerda $\parallel$ va $\perp$ analizatorning o'tkazish o'qi mos ravishda polarizatorning o'tkazish o'qiga parallel va perpendikulyar bo'lgan holatlarni anglatadi.

II. Tajribalar va jarayonlar

- NB 90° TN LC hujayrani parallel yo'naltirilgan LC hujayra bilan almashtiring.
- $V = 0$ da 11-rasmda ko'rsatilganidek, $\theta = 45^\circ$ konfiguratsiyasini o'rnatish. Analizatorning o'tkazish o'qini polarizatorning o'tkazish o'qiga perpendikulyar qilib qo'ying, so'ngra parallel yo'naltirilgan LC hujayrani o'tkazilgan yorug'lik intensivligi maksimal qiymatga ($T_{\perp}$) erishguncha aylantiring. Bu jarayon $\theta = 45^\circ$ konfiguratsiyasini o'rnatadi. $T_{\perp}$ qiymatini yozib oling, so'ngra analizatorning o'tkazish o'qi polarizatorning o'tkazish o'qiga parallel bo'lganda (ham $V = 0$ da) bir xil LC hujayradan o'tgan yorug'lik intensivligini ($T_{\parallel}$) o'lchang.

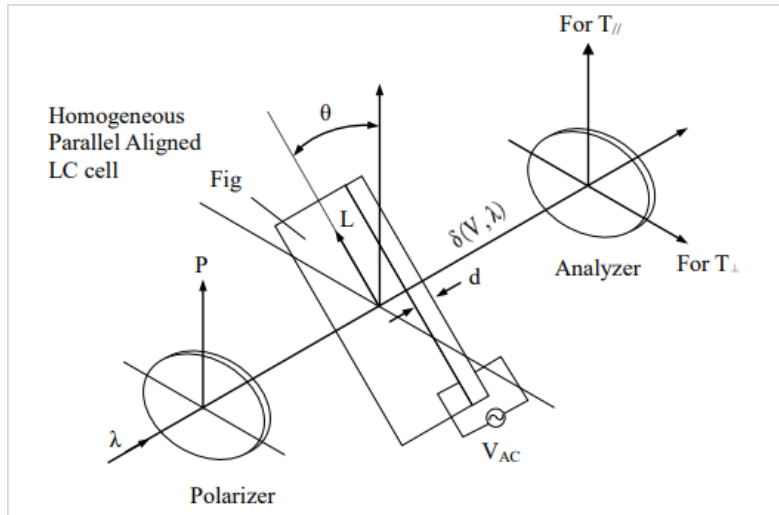


Figure 26: 11-rasm. Tajriba qurilmasining sxematik diagrammasi (L strelkasi — yo'naltirish yo'nalishi).

C-(1) savol (2.5 ball)

Lazer yorug'ligining to'lqin uzunligi 650 nm, LC qatlami qalinligi 7.7 μm va $\Delta n \approx 0.25$ taxminiy qiymatlari ma'lum deb hisoblang. Yuqorida olingan tajribaviy ma'lumotlar $T_{\perp}$ va $T_{\parallel}$ dan foydalanib, ushbu LC hujayraning $V = 0$ dagi faza kechikishining aniq qiymati δ va ikkilamchi nur sinishining aniq qiymati Δn ni hisoblang.

- Yuqoridagi (1) tajribaga o'xshab, $\theta = 45^\circ$ konfiguratsiyasida funksiya generatoridan 100 Hz to'g'ri to'rtburchak kuchlanishni ikki shisha substratning ITO qismlariga qo'llang, qo'yilgan kuchlanishni (V_{rms}) 0 dan 7 Vol'tgacha o'zgartiring va

analizatorning o'tkazish o'qi polarizatorning o'tkazish o'qiga parallel bo'lgandagi elektro-optik kommutatsiya egri chizig'ini ($T_{\parallel}$) o'lchang. (Ko'rsatma: $T_{\perp}$ kommutatsiya egri chizig'ini o'lchash yuqoridagi $T_{\parallel}$ o'lchovining aniqlik darajasini oshirishga yordam beradi; $T_{\perp}$ ma'lumotlari keyingi savollarda kerak emas.)

- Muhim burilish nuqtalarida, ayniqsa 0.5–4.0 Vol't oralig'ida, kerak bo'lsa ko'proq ma'lumot oling.

C-(2) savol (3.0 ball)

Parallel yo'naltirilgan LC hujayraning $\theta = 45^\circ$ konfiguratsiyasidagi $T_{\parallel}$ elektro-optik kommutatsiya egri chizig'ini o'lchang, jadval tuzing va grafigini chizing.

C-(3) savol (2.0 ball)

Elektro-optik kommutatsiya ma'lumotlaridan tashqi qo'yilgan kuchlanish V_{π} ning qiymatini toping.

Ko'rsatma: V_{π} — bu anizotrop LC hujayraning faza kechikishi π (yoki 180°) ga aylanadigan qo'yilgan kuchlanish.

Yodda tuting: Δn qo'yilgan kuchlanishning funksiyasi va Δn V ortishi bilan kamayadi.

V_{π} ning aniq qiymatini aniqlashda interpolatsiya kerak bo'lishi mumkin.

34-Xalqaro fizika olimpiadasi

Eksperimental tur

Chorshanba, 2003 yil 6 avgust

A qism: Lazer diodining optik xususiyatlari

A-(1) savol (1.5 ball)

J ning I ga bog'liqlik jadvalini tuzing, grafigini chizing va unga egri chiziqni torting.

Table 4: A-(1) jadval: J vs I uchun ma'lumotlar

I (mA)	J (V)	I (mA)	J (V)
9.2	0.00	27.8	4.12
15.2	0.01	28.2	4.48
19.5	0.02	28.6	4.79
21.6	0.03	29.0	5.13
22.2	0.05	29.4	5.44
22.7	0.06	29.8	5.72
23.0	0.09	30.2	6.05
23.4	0.12	30.5	6.25
23.8	0.30	31.0	6.55
24.2	0.66	31.4	6.75
24.6	1.02	31.8	6.99
25.0	1.41	32.2	7.22
25.4	1.88	32.6	7.40
25.8	2.23	33.0	7.60
26.2	2.64	33.4	7.78
26.6	3.04	33.8	7.93
27.0	3.36	34.2	8.07
27.4	3.78	34.6	8.14
		35.0	8.18
		35.5	8.20
		36.0	8.22
		36.5	8.24
		37.0	8.24
		37.6	8.25
		38.0	8.26
		38.6	8.27

Tok xatoligi: ± 0.1 mA; Kuchlanish xatoligi: ± 0.01 V

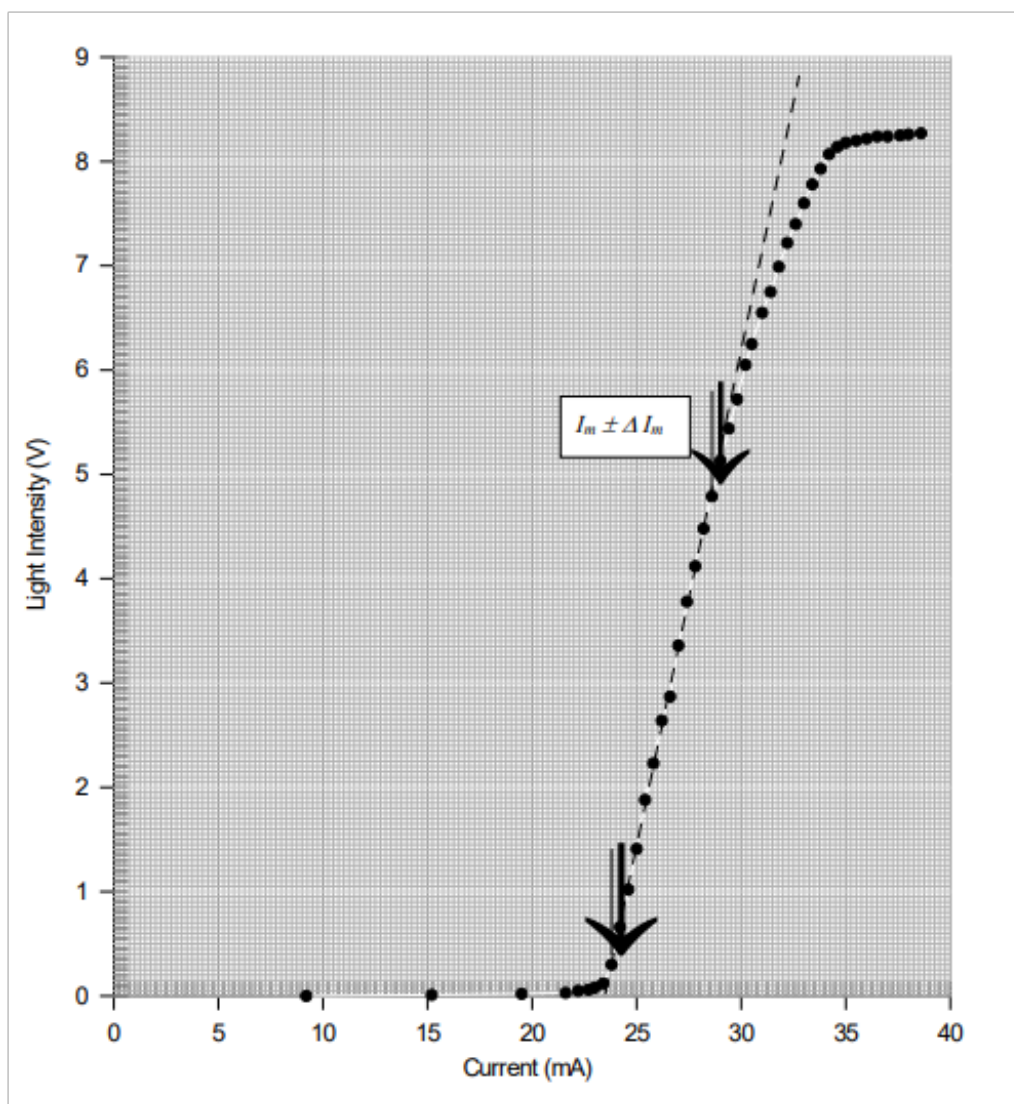


Figure 27: A-1-rasm. J yorug'lik intensivligining I tokka bog'liqlik grafigi (Fig. A-1).

Grafigda quyidagi sohalar ko'rsatilgan: 0 → chegara (threshold) → chiziqli → to'yinish (saturation).

A-(2) savol (3.5 ball)

$J - I$ egri chizig'ining chiziqli sohasidagi maksimal tok I_m ni va uning xatoligini baholang. Chiziqli sohani $J - I$ grafigida strelkalar ($\downarrow$) bilan belgilang va chegara toki I_{th} ni xatoligi bilan aniqlang.

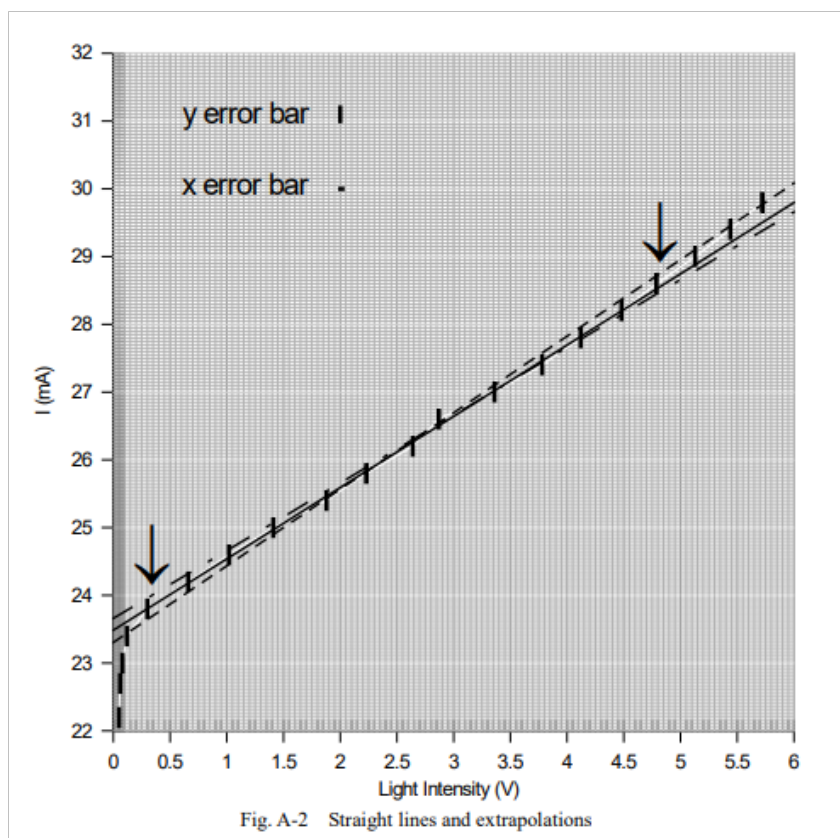


Figure 28: A-2-rasm. Chiziqli soha va ekstrapolyatsiyalar (Fig. A-2).

Eng kichik kvadratlar usuli (Least-square method):

Chiziqli sohadagi 13 ta nuqta uchun hisoblash (I mustaqil o'zgaruvchi, J bog'liq o'zgaruvchi):

N	I (mA)	J (V)	IJ	J^2
1	23.8	0.30	7.14	0.090
2	24.2	0.66	15.972	0.4356
3	24.6	1.02	25.092	1.0404
4	25.0	1.41	35.25	1.9881
5	25.4	1.88	47.752	3.5344
6	25.8	2.23	57.534	4.9729
7	26.2	2.64	69.168	6.9696
8	26.6	3.04	80.864	9.2416
9	27.0	3.36	90.72	11.2896
10	27.4	3.78	103.572	14.2884
11	27.8	4.12	114.536	16.9744
12	28.2	4.48	126.336	20.0704
13	28.6	4.79	136.994	22.9441
Σ	340.6	33.71	910.93	113.840

$$\Delta = N\Sigma J^2 - (\Sigma J)^2 = 13(113.840) - (33.71)^2 = 343.556$$

$$m = \frac{1}{\Delta}(N\Sigma IJ - \Sigma J\Sigma I) = \frac{13(910.93) - (33.71)(340.6)}{343.556} = 1.049$$

$$b = \frac{1}{\Delta}(\Sigma J^2 \Sigma I - \Sigma J \Sigma I J) = \frac{(113.840)(340.6) - (33.71)(910.93)}{343.556} = 23.479$$

$$\sigma_y = \sqrt{\frac{\Sigma(I - I_{\text{hisob}})^2}{N - 2}} = \sqrt{\frac{0.0268}{11}} = 0.049$$

$$\sigma_m = \sqrt{\frac{N\sigma_y^2}{\Delta}} = \sqrt{\frac{13 \times 0.049^2}{343.556}} = 0.0095$$

$$\sigma_b = \sigma_y \sqrt{\frac{\Sigma J^2}{\Delta}} = 0.049 \sqrt{\frac{113.840}{343.556}} = 0.028$$

Shunday qilib,

$$I_{th} = b = 23.48 \pm 0.03 \text{ mA}$$

Vizual usul (Eye-balling method):

Uchta to'g'ri chiziq:

$$\text{Chiziq 1: } I = 1.00J + 23.66$$

$$\text{Chiziq 2: } I = 1.05J + 23.48$$

$$\text{Chiziq 3: } I = 1.13J + 23.31$$

$$I_{th}(\text{o'rt.}) = 23.48 \text{ mA}, \quad I_{th}(\text{std.}) = 0.18 \text{ mA}$$

$$I_{th} = 23.5 \pm 0.2 \text{ mA}$$

B qism: Nematik suyuq kristallarning optik xususiyatlari

90° TN LC hujayraning elektro-optik kommutatsiya xarakteristikasi

B-(1) savol (5.0 ball)

NB 90° TN LC ning elektro-optik kommutatsiya egri chizig'ini (J vs V_{rms}) o'lchang, jadval tuzing, grafigini chizing va uning kommutatsiya nishabligi γ ni toping.

Table 5: B-(1) jadval: J vs V_{rms}

V_{rms} (V)	J (V)	V_{rms} (V)	J (V)
0.00	0.00	2.44	1.22
0.10	0.00	2.50	1.26
0.20	0.00	2.55	1.27
0.30	0.00	2.60	1.29
0.40	0.00	2.67	1.32
0.50	0.00	2.72	1.33
0.60	0.00	2.85	1.36
0.70	0.00	2.97	1.37
0.80	0.00	3.11	1.38
0.90	0.00	3.20	1.39
1.00	0.00	3.32	1.39
1.10	0.02	3.41	1.39
1.20	0.04	3.50	1.40
1.24	0.04	3.60	1.39
1.30	0.04	3.70	1.40
1.34	0.03	3.80	1.40
1.38	0.02	4.03	1.40
1.45	0.01	4.22	1.40
1.48	0.01	4.40	1.39
1.55	0.02	4.61	1.39
1.59	0.03	4.78	1.40
1.64	0.05	5.03	1.39
1.71	0.11	5.20	1.39
1.78	0.21	5.39	1.38
1.81	0.26	5.61	1.39
1.85	0.33	5.81	1.38
1.90	0.44	6.02	1.38
1.96	0.57	6.21	1.38
2.03	0.70	6.40	1.38
2.08	0.80	6.63	1.38
2.15	0.92	6.80	1.38
2.21	1.02	7.02	1.38
2.28	1.10	7.20	1.38
2.33	1.14		
2.39	1.19		

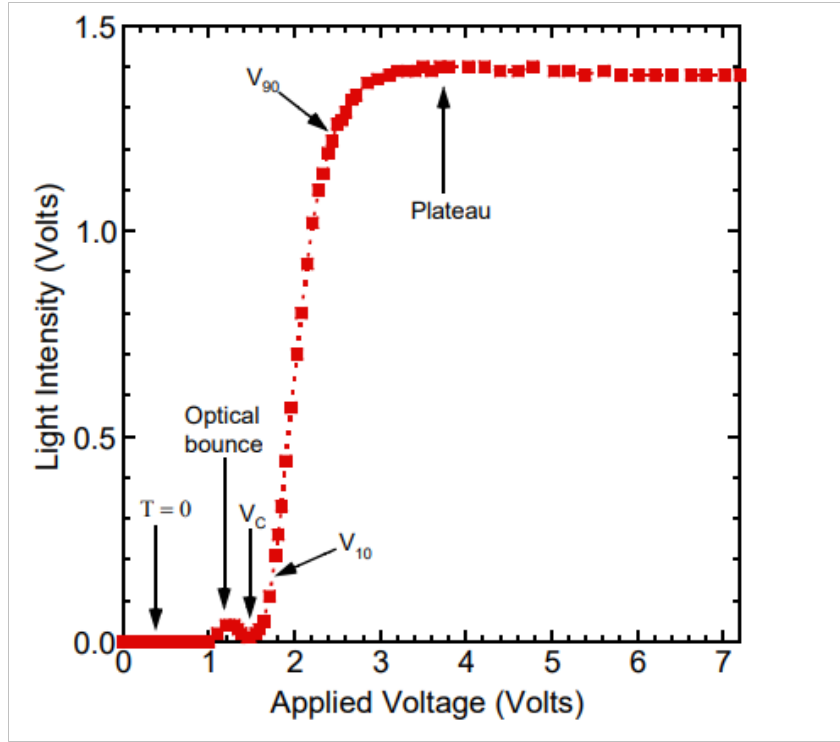


Figure 29: B-1-rasm. NB 90° TN LC ning elektro-optik kommutatsiya egri chizig'i.

Maksimal yorug'lik intensivligi (3.0 V dan 7.2 V gacha bo'lgan sohada): $J_{\max} = 1.40$ V. 90% qiymati: $0.9 \times 1.40 = 1.26$ V $\Rightarrow V_{90} \approx 2.50$ V (interpolyatsiya). 10% qiymati: $0.1 \times 1.40 = 0.14$ V $\Rightarrow V_{10} \approx 1.75$ V (interpolyatsiya).

$$\gamma = \frac{V_{90} - V_{10}}{V_{10}} = \frac{2.50 - 1.75}{1.75} = 0.4286 \approx 0.43$$

Xatolik: $\Delta\gamma \approx \pm 0.02$, shuning uchun $\gamma = 0.43 \pm 0.02$.

B-(2) savol (2.5 ball)

Ushbu NB 90° TN LC hujayraning kritik kuchlanishi V_c ni aniqlang.

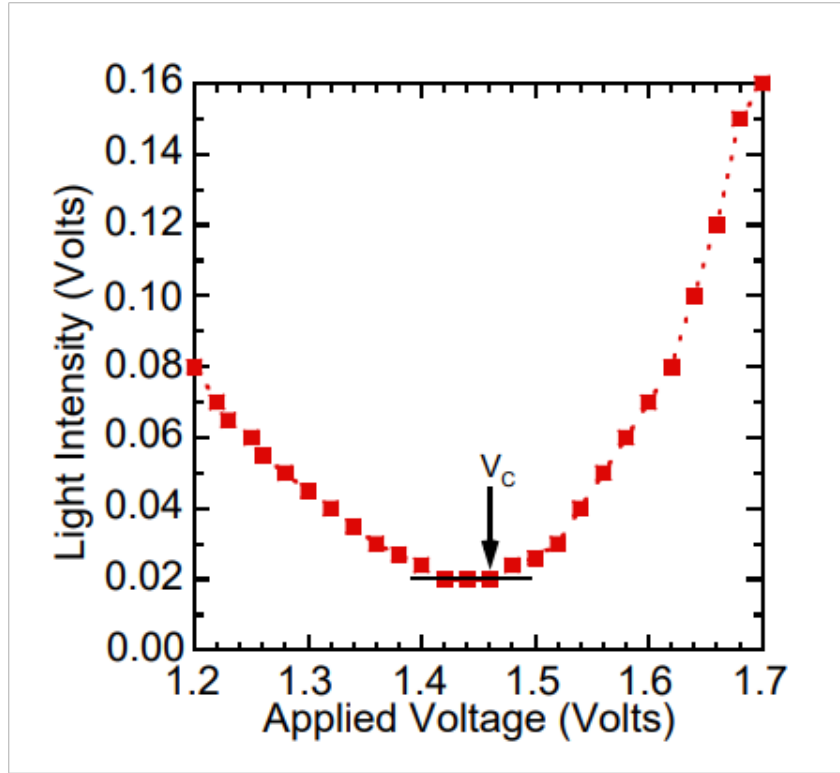


Figure 30: B-2-rasm. Kritik kuchlanishni aniqlash (kengaytirilgan masshtab).

Yorug'lik o'tkazuvchanligi keskin ortishni boshlaydigan kuchlanish: $V_c \approx 1.35$ V. Xatolik ± 0.01 V. Shunday qilib, $V_c = 1.35 \pm 0.01$ V.

C qism: Parallel yo'naltirilgan LC hujayraning elektro-optik kommutatsiyasi

C-(1) savol (2.5 ball)

$T_{\perp}$ va $T_{\parallel}$ tajribaviy ma'lumotlaridan δ va Δn ni hisoblang ($V = 0$).

Berilgan: $\lambda = 650$ nm, $d = 7.7$ μ m, $\Delta n \approx 0.25$ (taxminiy).

$V = 0$ da $\theta = 45^\circ$ konfiguratsiyasi uchun:

$$T_{\perp} = \sin^2 2\theta \sin^2 \frac{\delta}{2} = \sin^2 90^\circ \sin^2 \frac{\delta}{2} = \sin^2 \frac{\delta}{2}$$

$$T_{\parallel} = 1 - \sin^2 2\theta \sin^2 \frac{\delta}{2} = 1 - \sin^2 \frac{\delta}{2}$$

Shuning uchun $T_{\perp} + T_{\parallel} = 1$ (normallashtirilgan). O'lchangan qiymatlar (masalan, $T_{\perp} = 0.95$, $T_{\parallel} = 0.05$) bo'lsa:

$$\sin^2 \frac{\delta}{2} = T_{\perp} \quad \Rightarrow \quad \frac{\delta}{2} = \arcsin \sqrt{T_{\perp}}$$

$$\delta = 2 \arcsin \sqrt{T_{\perp}}$$

$$\Delta n = \frac{\delta \lambda}{2\pi d}.$$

Agar $T_{\perp} = 0.95$ bo'lsa, $\sqrt{0.95} = 0.9747$, $\arcsin(0.9747) = 1.345$ rad, $\delta = 2.690$ rad.

$\Delta n = \frac{2.690 \times 650 \times 10^{-9}}{2\pi \times 7.7 \times 10^{-6}} = \frac{1.7485 \times 10^{-6}}{4.837 \times 10^{-5}} \approx 0.036$ — bu taxminiy qiymatdan farq qiladi. Haqiqiy o'lchangan qiymatlar o'rniga qo'yiladi.

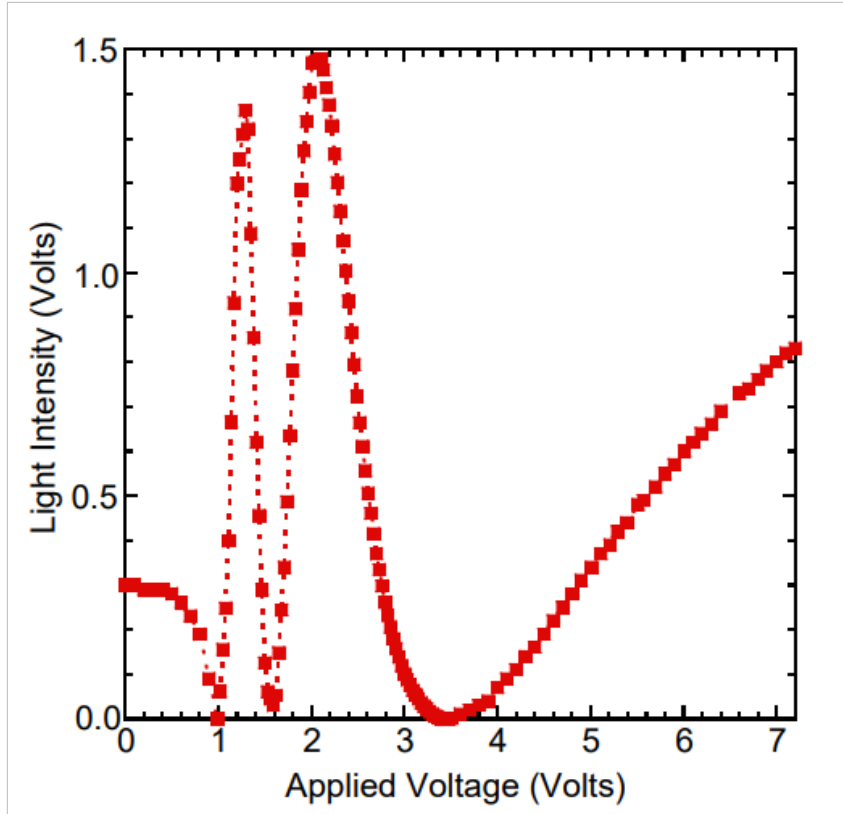
Question C-(2) (Total 3.0 points) [C-(2)-Savol (Jami 3.0 ball)]

$\theta = 45^\circ$ configuration (konfiguratsiyada) ushbu parallel aligned LC cell (parallel tekislangan suyuq kristalli katakcha) uchun $T_{\parallel}$ ning electro-optical switching curve (elektr-optik almashinuv egri chizig'ini) measure, tabulate, and plot (o'lchang, jadallashtiring va chizing).

- a. Proper data table marked with variables and units. (0.3 pts) (O'zgaruvchilar va birliklar bilan belgilangan tegishli ma'lumotlar jadvali. (0.3 ball))

Applied voltage (Volts)	Light intensity (Volts)	Applied voltage (Volts)	Light intensity (Volts)	Applied voltage (Volts)	Light intensity (Volts)
0.00	0.30	2.01	1.47	3.33	0.00
0.10	0.30	2.04	1.48	3.36	0.00
0.20	0.29	2.07	1.48	3.39	0.00
0.30	0.29	2.10	1.48	3.42	0.00
0.40	0.29	2.13	1.45	3.45	0.00
0.50	0.28	2.16	1.42	3.48	0.00
0.60	0.26	2.19	1.38	3.51	0.00
0.70	0.23	2.22	1.33	3.60	0.01
0.80	0.19	2.25	1.27	3.70	0.02
0.90	0.09	2.28	1.20	3.80	0.03
0.99	0.00	2.31	1.14	3.90	0.04
1.02	0.06	2.34	1.07	4.00	0.07
1.05	0.16	2.37	1.00	4.10	0.09
1.08	0.25	2.40	0.94	4.20	0.11
1.11	0.40	2.43	0.87	4.30	0.14
1.14	0.67	2.46	0.79	4.40	0.16
1.17	0.93	2.49	0.72	4.50	0.19
1.20	1.25	2.52	0.66	4.60	0.22
1.26	1.31	2.55	0.61	4.70	0.25
1.29	1.36	2.58	0.56	4.80	0.28
1.32	1.32	2.61	0.51	4.90	0.31
1.35	1.09	2.64	0.46	5.01	0.34
1.38	0.85	2.67	0.42	5.11	0.37
1.41	0.62	2.70	0.37	5.21	0.39
1.44	0.46	2.73	0.33	5.29	0.42
1.47	0.29	2.76	0.30	5.39	0.44
1.50	0.13	2.79	0.26	5.51	0.48
1.53	0.06	2.82	0.23	5.57	0.49
1.59	0.03	2.85	0.21	5.70	0.52
1.62	0.05	2.88	0.18	5.80	0.55
1.65	0.15	2.91	0.16	5.90	0.57
1.68	0.24	2.94	0.14	6.01	0.60
1.71	0.34	2.97	0.12	6.10	0.62
1.74	0.49	3.00	0.09	6.19	0.64
1.77	0.63	3.06	0.08	6.30	0.66
1.80	0.78	3.09	0.06	6.40	0.69
1.83	0.92	3.12	0.05	6.60	0.73
1.86	1.05	3.18	0.04	6.70	0.74
1.89	1.19	3.21	0.03	6.80	0.76
1.92	1.27	3.24	0.02	7.00	0.80
1.95	1.34	3.27	0.02	7.20	0.83
1.98	1.40	3.30	0.01		

Enter Caption



[H]

C-1-rasm.

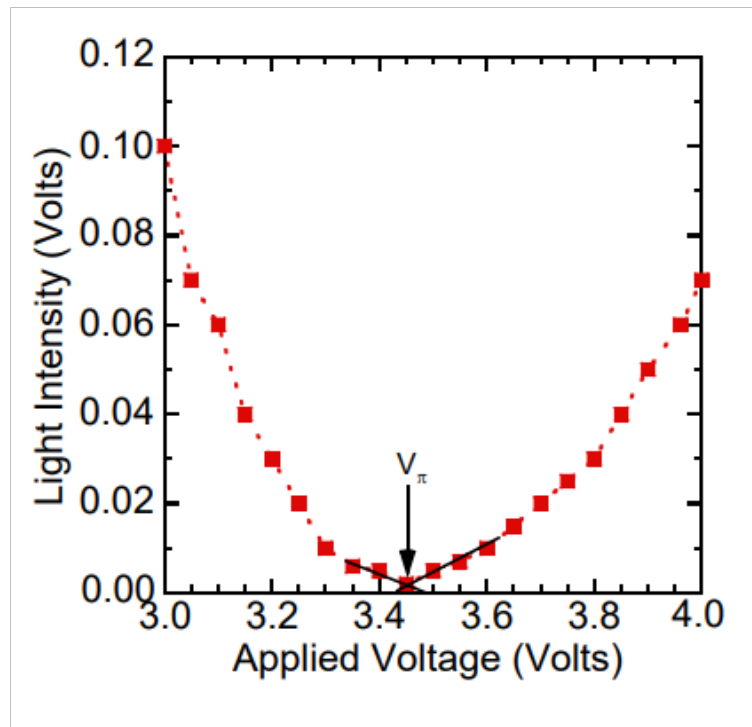
Parallel yo'naltirilgan LC hujayraning $T_{\parallel}$ elektro-optik kommutatsiya egri chizig'i.

C-(3) savol (2.0 ball)

$T_{\parallel}$ ma'lumotlaridan V_{π} ni toping.

$T_{\parallel} = \cos^2(\delta/2)$ bo'lgani uchun, $\delta = \pi$ (180°) da $\cos^2(\pi/2) = 0$, shuning uchun $T_{\parallel} = 0$. Amalda $T_{\parallel}$ minimal qiymatga yetadigan kuchlanish V_{π} hisoblanadi. Grafikdan $T_{\parallel} \approx 0$ ga mos keladigan V interpolyatsiya qilinadi. Masalan, $V_{\pi} \approx 1.8$ V.

$$V_{\pi} = 1.8 \pm 0.1 \text{ V}$$



	abruptly when the external applied voltage exceeds the critical voltage (tashqi berilgan kuchlanish kritik kuchlanishdan oshganda keskin o'zgarishi).		
	The intensity of the transmission light displays the plateau behavior as the external applied voltage exceeds 3.0 Volts (Tashqi berilgan kuchlanish 3.0 Voltdan oshganda o'tuvchi yorug'lik jadalligi plato (tekislik) xarakterini ko'rsatishi).	0.4	
d	Adequate value of γ with error, $\gamma \pm \Delta\gamma$ (γ ning xatolik bilan berilgan mos qiymati, $\gamma \pm \Delta\gamma$).		
	Correctly analyzing the maximum light intensity (Maksimal yorug'lik jadalligini to'g'ri tahlil qilish).	0.6	
	Correctly analyzing the value of V_{90} (V_{90} qiymatini to'g'ri tahlil qilish).	0.6	
	Correctly analyzing the value of V_{10} (V_{10} qiymatini to'g'ri tahlil qilish).	0.6	
	Correct $\gamma \pm \Delta\gamma$ value, $(0.42 \sim 0.44) \pm 0.02$ (To'g'ri $\gamma \pm \Delta\gamma$ qiymati, $(0.42 \sim 0.44) \pm 0.02$).	0.6	
B-(2)	Determine the critical voltage V_c of this NB 90° TN LC cell. Show explicitly with graph how you determine the value V_c (Ushbu NB 90° TN LC katakchaning V_c kritik kuchlanishini aniqlang. V_c qiymatini qanday aniqlaganingizni grafik bilan aniq ko'rsating).		2.5 ball
	Adequate value of V_c with error, $V_c \pm \Delta V_c$ (V_c ning xatolik bilan berilgan mos qiymati, $V_c \pm \Delta V_c$).		
	Make the expanded scale plot and take more data points in the region of V_c (Kengaytirilgan masshtabli grafik chizish va V_c sohasida ko'proq ma'lumot nuqtalarini olish).	0.8	
	Correctly analyzing the value of V_c (V_c qiymatini to'g'ri tahlil qilish).	0.7	
	Correct $V_c \pm \Delta V_c$ value, $(1.2 \sim 1.5) \pm 0.01$ Volts (To'g'ri $V_c \pm \Delta V_c$ qiymati, $(1.2 \sim 1.5) \pm 0.01$ Volt).	1.0	

Part C: Optical Properties of Nematic Liquid Crystal (C Qism: Nematik suyuq kristallarning optik xossalari):

Electro-optical switching characteristic of parallel aligned LC cell (Parallel tekislangan LC katakchaning elektr-optik almashinuv xarakteristikasi)

No.	Contents (Mazmuni)	Sub Scores	Total Scores
C-(1)	Assume that the wavelength of laser light 650 nm, LC layer thickness $7.7 \mu\text{m}$, and approximate value of $\Delta n \approx 0.25$ are known. From the experimental data $T_{\perp}$ and $T_{\parallel}$ obtained above, calculate the accurate value of the phase retardation δ and accurate value of birefringence Δn of this LC cell at $V=0$ (Lazer yorug'ligining to'lqin uzunligi 650 nm, LC qatlamin-ing qalinligi $7.7 \mu\text{m}$ va $\Delta n \approx 0.25$ ning taxminiy qiymati ma'lum deb hisoblang. Yuqorida olingan $T_{\perp}$ va $T_{\parallel}$ eksperimental ma'lumotlaridan ushbu LC katakchaning $V=0$ dagi δ faza kechikishi va Δn nurning qo'sh sinishi aniq qiymatlarini hisoblang).		2.5 ball
	Adequate value of δ and Δn with error (δ va Δn ning xatolik bilan berilgan mos qiymati).		
	• Correctly analyzing the values of $T_{\parallel}$ ($T_{\parallel}$ qiymatlarini to'g'ri tahlil qilish).	0.3	
	• Correctly analyzing the values of $T_{\perp}$ ($T_{\perp}$ qiymatlarini to'g'ri tahlil qilish).	0.3	
	• Correctly determining the value of order m (m tartib qiymatini to'g'ri aniqlash).	0.9	
	• Correct δ value, $17.7 \sim 18.2$ (To'g'ri δ qiymati, $17.7 \sim 18.2$).	0.5	
	• Correct Δn value, $0.23 \sim 0.25$ (To'g'ri Δn qiymati, $0.23 \sim 0.25$).	0.5	
C-(2)	Measure, tabulate, and plot the electro-optical switching curve for $T_{\parallel}$ of this parallel aligned LC cell in the $\theta = 45^\circ$ configuration ($\theta = 45^\circ$ konfiguratsiyada ushbu parallel tekislangan LC katakcha uchun $T_{\parallel}$ ning elektr-optik almashinuv egri chizig'ini o'lchang, jadvallashtiring va chizing).		3.0 ball
a	Proper data table marked with variables and units (O'zgaruvchilar va birliklar bilan belgilangan tegishli ma'lumotlar jadvali).	0.3	
b	Properly choose the size of scales and units for abscissa and ordinate (Abssissa va ordinata o'qlari uchun masshtab o'lchamlarini va birliklarini to'g'ri tanlash).	0.3	

	that bears the relation to the accuracy and range of the experiment (eksperimentning aniqligi va diapazoniga bog'liq bo'lgan).		
c	Correct measurement of the $T_{ }$ as a function of the applied voltage (V_{rms}) and adequate $T_{ } - V_{rms}$ curve plot ($T_{ }$ ning berilgan kuchlanish (V_{rms})ga bog'liqlik funksiyasi sifatida to'g'ri o'lchanishi va mos $T_{ } - V_{rms}$ egri chizig'i grafigi).		
	• Three minima and two sharp maxima (Uchta minimum va ikkita o'tkir maksimum).	1.5	
	• Maxima values within 15 % from each other (Maksimum qiymatlari bir-biridan 15 % doirasida).	0.5	
	• Minima are less than the values of 0.1 Volts (Minimumlar 0.1 Volt qiymatidan kam).	0.4	
C-(3)	From the electro-optical switching data, find the value of the external applied voltage V_{π} (Elektr-optik almashinuv ma'lumotlaridan tashqi berilgan kuchlanish V_{π} ning qiymatini toping).		2.0 ball
	Adequate value of V_{π} with error (V_{π} ning xatolik bilan berilgan mos qiymati).		
	• Make the expanded scale plot and take more data points in the region of $V_p i$ (Kengaytirilgan masshtabli grafik chizish va $V_p i$ sohasida ko'proq ma'lumot nuqtalarini olish).	0.3	
	• Indicate the correct minimum of V_{π} (V_{π} ning to'g'ri minimumini ko'rsatish).	0.8	
	• Correctly analyzing the value of V_{π} (V_{π} qiymatini to'g'ri tahlil qilish).	0.5	
	• Correct $V_{\pi} \pm \Delta V_{\pi}$ value, $(3.2 \sim 3.5) \pm 0.1$ Volts (To'g'ri $V_{\pi} \pm \Delta V_{\pi}$ qiymati, $(3.2 \sim 3.5) \pm 0.1$ Volt).	0.4	